Highlights

Auditing the Decision Value of Physics-Constrained Wind Power Forecasting

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- Scoping audit of 46 physics-constrained wind forecasting studies reveals 95.7% open-code deficit.
- Audited literature exhibits zero documented failure regimes, boundary conditions, or trade-offs.
- Code audit of PCT identifies inverted divergence arguments altering loss gradient scaling.
- Scrambled prior negative controls isolate genuine physical guidance from numerical smoothing.
- Degenerate zero-bid diagnostic shows standalone balancing cashflows decouple from forecast skill.

Auditing the Decision Value of Physics-Constrained Wind Power Forecasting

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Abstract

Reductions in point forecasting error from physical constraints do not establish that empirical gains stem from valid physical information, nor do they guarantee operational decision value. In the published literature, physical constraints are widely credited with improving renewable energy forecasts, but existing evaluation protocols rarely isolate domain physical guidance from parametric capacity, audit software execution fidelity, or measure asymmetric operational costs. We audit the attributional foundation of physics-constrained wind power forecasting through systematic literature scoping ($N = 46$), static software code inspection, and prospective controlled ablations across three commercial UK wind farms. Within the scoped cohort (2023–2026), 0.0% of studies document operational failure boundaries, 95.7% lack public source code, and 50.0% evaluate constraints via confounded module excisions. Software inspection of the open-source PCT benchmark uncovers an inverted argument in its divergence loss that induces an analytical gradient singularity, though its bivariate kernel coupling remains mathematically sound. In controlled experiments holding the neural backbone and post-hoc quantile calibration constant, strong empirical power curve constraints ($\lambda = 1.0$) reduce normalized asymmetric decision loss by 1.22% under sample restriction (Penmanshiel), whereas scale-matched scrambled priors inflate loss by 68.1%, proving prior specificity under data scarcity. Conversely, weak penalties ($\lambda = 0.01$) mimic non-specific numerical smoothing, while offline historical weather proxy staleness ($k > 12$) monotonically degrades performance across all sites. Finally, degenerate zero-bid probing demonstrates that standalone unhedged balancing cashflows decouple from forecast quality under two-settlement markets, indicating that verifiable physical forecasting claims require backbone-matched negative controls, disclosed data budgets, and decision-aligned evaluation.

Keywords: Physics-Constrained Deep Learning, Wind Power Forecasting, Prediction-Decision Decoupling, Conditional Inductive Priors, Systematic Audit, Implementation Fidelity, Machine Learning Reproducibility

1. Introduction

In electric power systems and wholesale markets, the operational value of wind power forecasting is determined by asymmetric decision losses in forward scheduling and balancing settlement [1, 2]. Under standard balancing rules, shortfall penalties for under-delivery often differ substantially from spill revenues for over-delivery. In these asymmetric operational regimes, conventional symmetric regression metrics, such as Root Mean Squared Error (RMSE) and Mean Absolute Error (MAE), fail to reflect downside financial and system reliability exposure [3, 4]. To improve generalization and enforce physically realistic outputs, an expanding literature embeds aerodynamic limits, empirical power curves, and kinematic equations directly into deep neural network architectures and loss functions [5, 6, 7, 8]. A foundational premise underpins this research paradigm: that reported reductions in forecasting error can be directly attributed to the valid physical domain knowledge encoded within the constraint.

However, prevailing reporting conventions, software availability deficits, and ablation designs obscure this attributional premise. First, contemporary studies predominantly document positive outcomes across nominal test conditions; empirical failure regimes, operational trade-offs, and constraint breakdowns remain largely undocumented. Second, public source code availability remains rare in this domain, leaving the mathematical fidelity of published loss formulations and computational graph executions largely uninspected [9, 10]. Third, when ablation studies are reported, they frequently excise multi-layer neural submodules (such as graph neural net-

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work branches or attention blocks) rather than toggling the constraint penalty while holding network capacity constant. Such architectural ablations inevitably confound aerodynamic inductive bias with parametric representation capacity. Consequently, the literature leaves central empirical questions unresolved: Does a physical penalty succeed because the network internalizes genuine aerodynamic relationships, or because the penalty merely acts as a non-specific numerical regularizer? Under what training sample budgets, temporal proxy delays, and evaluation metrics do these empirical advantages persist, diminish, or invert?

To examine the attributional basis of physics-constrained gains, we conduct an empirical audit across three commercial UK wind farms (Kelmarsh, Hill of Towie, and Penmanshiel). Holding the forecasting backbone and post-hoc quantile calibration constant, we introduce scale-matched pseudo-scrambled priors as negative controls, systematically perturb prior parameters, and evaluate model behavior across sample sizes. The controlled experiments reveal that the decision value of physical constraints is contingent on data volume, prior alignment, and metric selection:

1. **Prior Specificity under Sample Scarcity**: On Penmanshiel under restricted training data ($N_{\text{train}} = 1{,}583$), embedding an authentic empirical power curve prior reduces normalized asymmetric decision loss by 1.22%. Conversely, permuting the wind-to-power mapping via a pseudo-scrambled control inflates decision loss by 68.1%, demonstrating prior specificity under strong regularization in low-sample regimes.

2. **Diminishing Advantages under Sample Expansion**: When evaluated across training sample budgets, fixed historical empirical priors provide modest dispatch cost reductions under constrained training windows ($N_{\text{train}} \leq 1{,}000$). However, because the prior is estimated from the full pre-cutoff historical record, it introduces an auxiliary seasonal data budget. As the model training window expands to $N_{\text{train}} = 5{,}000$, this prior advantage diminishes and slightly reverses, where rigid historical curves constrain the model from fitting shifting local dynamics.

3. **Software Implementation Divergence and Settlement Decoupling**: Computational graph inspection of an open-source constraint implementation reveals an inverted parameter argument in Jensen–Shannon divergence loss, altering backpropagated gradient scales by approximately 2.5 times. Downstream, market settlement analysis demonstrates that standalone unhedged balancing cashflows decouple from forecast quality: a degenerate zero-bid baseline ($\hat{y} \equiv 0$) mechanically generates six-figure apparent cashflows purely through uncontracted physical spill, confirming that single-settlement accounting cannot serve as a reliable proxy for forecasting skill.

1.1. Summary of Contributions

This investigation addresses the attribution and evaluation of physics-constrained forecasting through three interconnected contributions:

- **Systematic Scoping Audit of Literature and Code Availability**: We evaluate 46 contemporary studies (2023–2026) under PRISMA-ScR guidelines, quantifying the reporting absence of operational failure regimes (0/46), the open-code deficit (44/46), and the prevalence of structural module-level ablation confounding (23/46).

- **Backbone-Matched Controlled Empirical Ablation**: By pairing authentic priors with scrambled negative controls across multi-seed evaluations on three commercial wind farms, we isolate physical inductive bias from non-specific numerical smoothing. We delineate the operational boundaries under which physical constraints assist, become indistinguishable from random regularization, or degrade downstream decision performance.

- **Computational Graph Audit and Settlement Decoupling Diagnostics**: We uncover implementation divergence in public constraint codebases and demonstrate why unhedged single-settlement imbalance cashflows decouple from forecast quality, establishing an evaluation framework anchored to normalized asymmetric decision loss.

Empirical Scope and Methodological Boundaries. This investigation operates within four defined empirical boundaries: first, the literature audit evaluates reporting and ablation practices across the 46 studies identified under our prespecified PRISMA-ScR protocol (38 journal articles and 8 preprints, 2023–2026), characterizing reporting patterns within this scoping cohort rather than asserting universal claims across unbounded keyword taxonomies; second, the software audit inspects the computational graph of the open-source PCT implementation; third, empirical ablations evaluate boundary behaviors across three commercial UK wind farms under matched DLinear backbones, with PatchTST serving as an architectural sensitivity probe; and fourth, downstream evaluations analyze operational decision risk and

single-settlement imbalance cashflow diagnostics under UK BSC rules, characterizing metric failure modes rather than full commercial portfolio clearing.

2. Related Work and Positioning

To position this scoping audit within computational energy systems and empirical machine learning methodology, we review three related literatures: (i) physics-constrained formulations in renewable energy forecasting, (ii) operational decision evaluation in wind power prediction, and (iii) reproducibility, reporting standards, and software auditing in scientific machine learning.

2.1. Physical Constraints in Wind Power Forecasting

Embedding physical domain knowledge into neural networks aims to combine governing physical principles with data-driven function approximation [6, 5]. In wind power forecasting, these constraints typically follow three distinct technical formulations:

Operational Envelope Bounds. Enforcing aerodynamic operating thresholds (such as cut-in, rated, and cut-out wind speeds, or theoretical Betz envelopes) via bounded activation functions or post-hoc projections guarantees that outputs respect physical capacity boundaries [11]. However, hard clipping can alter gradient flows near boundary cutoffs. When turbulent boundary-layer conditions push turbine operations near rated capacity, zero-gradient saturation regions can impede gradient descent, restricting local adaptation to rapid meteorological transitions.

Kinematic Wake Formulations. Spatio-temporal models commonly embed steady-state kinematic wake equations—such as the 1D Jensen or Bastankhah–Porté-Agel Gaussian deficit models [7, 12]—into Graph Neural Network (GNN) adjacency matrices. While this approach incorporates structural layout geometry, substituting static engineering approximations for dynamic Navier–Stokes fluid dynamics can constrain network representations during complex convective conditions. Furthermore, published studies frequently attribute performance gains to the embedded physical formulation without isolating whether the expressive capacity of the GNN backbone alone explains the improvement.

Distributional Regularization. Penalizing deviations from historical joint wind speed–power distributions via a statistical divergence objectives (such as Wasserstein distance or Jensen–Shannon divergence [8, 11]) introduces additional optimization dynamics [13]. Static distribution alignment regularizes predictions against precomputed empirical manifolds. When meteorological regimes shift or input observations suffer from temporal lag, divergence penalties can penalize genuine operational transients, creating optimization trade-offs between distribution fidelity and immediate dispatch accuracy.

2.2. Decision-Theoretic Evaluation and Settlement Mechanics

Wind power forecasting models have traditionally been evaluated using symmetric statistical error metrics, such as MAE, RMSE, and R^2 [14, 15]. Madsen et al. [14] established standardized verification protocols highlighting the importance of evaluating models across horizons, operational regimes, and normalized capacity metrics. Optis et al. introduced the *WE-Validate* framework [15], emphasizing distributionally resistant metrics tailored to complex atmospheric conditions.

However, power system operations demonstrate that statistical error reductions do not guarantee economic value under asymmetric market penalties [2, 16]. Pinson et al. [4] demonstrated that optimal bidding under asymmetric imbalance penalties is governed by conditional quantiles matching the cost ratio, highlighting the operational decoupling between point error and decision risk. Downstream grid dispatch penalizes reserve shortfalls far more severely than excess generation [17, 18]. Furthermore, real-world balancing arrangements—such as the UK Elexon Balancing and Settlement Code (BSC)—net energy deviations over discrete 30-minute settlement periods. In isolated single-settlement accounting, unhedged cashflows can create apparent monetary surpluses through systematic negative bias, decoupling cashflow from forecasting skill. Evaluating forecasters strictly on uncalibrated point errors or unhedged cashflows obscures their true operational efficacy.

2.3. Reproducibility Standards and Software Auditing in Scientific AI

Empirical machine learning has faced increasing scrutiny over data leakage, underspecification, and implementation discrepancies [19, 20, 9, 3]. Kapoor and Narayanan [9] documented how subtle data leakage vectors undermine reproducibility across seventeen scientific domains. Gundersen and Kjensmo [20] evaluated 400 papers from major AI conferences, finding that fewer than 20% shared executable source code or reproducible experimental protocols.

To address these vulnerabilities, procedural frameworks have gained traction. Pineau et al. [10] developed the NeurIPS Machine Learning Reproducibility Checklist, demonstrating that explicit reporting mandates improve disclosure of code, data, hyperparameter choices, and error bounds. D'Amour et al. [19] showed that models with equivalent in-distribution test accuracy often diverge under distribution shift due to underspecified training criteria. Concurrently, Lipton and Steinhardt [3] analyzed recurring pitfalls in empirical scholarship: confusing empirical association with theoretical explanation, failing to isolate independent sources of performance gains, and employing decorative mathematics that does not govern empirical outcomes.

While general machine learning venues have increasingly adopted these auditing norms, physics-constrained energy forecasting has developed largely without standardized reporting controls. As summarized in Table 1, existing reviews focus primarily on model typologies, rarely evaluating constraint provenance, software execution fidelity, or ablation confounding. This investigation operationalizes these empirical standards through systematic auditing of lifecycle provenance, software parity, and backbone-matched controlled ablations.

3. Taxonomic Framework: The Five-Dimensional Constraint Lifecycle

In renewable energy forecasting, designating a neural network as "physics-constrained" often groups disparate mathematical, structural, and empirical operations under a single umbrella term [3]. To establish an operational foundation for systematic auditing, we define a structured **Constraint Lifecycle Taxonomy**.

3.1. Unit of Analysis and Lifecycle Representation

In our scoping framework, the primary unit of analysis is the individual **forecasting study** ($N = 46$ unique scoping studies), characterized by its primary operational constraint mechanism. When a study incorporates multiple physical elements (e.g., combining an empirical power curve regularizer with a spatial wake graph architecture), it is classified according to the dominant physical constraint governing the computational graph, while auxiliary components are documented in the supplementary ledger. Formally, we characterize the primary constraint architecture C of each audited study as a 5-tuple operating across orthogonal lifecycle stages:

$$C = \langle \mathcal{D}_1, \mathcal{D}_2, \mathcal{D}_3, \mathcal{D}_4, \mathcal{D}_5 \rangle, \tag{1}$$

where $\mathcal{D}_1$ represents parameter provenance, $\mathcal{D}_2$ denotes application stage, $\mathcal{D}_3$ specifies inference telemetry requirements, $\mathcal{D}_4$ captures implementation software fidelity, and $\mathcal{D}_5$ evaluates ablation semantics and harm reporting. The lifecycle trajectory from physical specification to operational evaluation is illustrated in Figure 1.

3.2. Operational Definitions of Lifecycle Dimensions

The taxonomy evaluates each audited constraint architecture across five orthogonal dimensions:

Dimension 1: Parameter Provenance ($\mathcal{D}_1$). Parameter provenance identifies the epistemic origin and calibration basis of the numerical values governing a constraint:

- *Intrinsic constraints*: parameters derive analytically from governing physical invariants, manufacturer nameplate limits, or classical equations without empirical fitting (e.g., the Betz limit $C_p \leq 16/27$ or kinematic wake equations).
- *Historical-static priors*: parameters or reference distributions are estimated offline from historical operational SCADA archives and held fixed during inference (e.g., empirical power curves fitted via kernel density estimation).
- *Learned-online constraints*: parameters adapt dynamically during test-time inference via an auxiliary neural subnetwork.

When a study omits calibration details, it is designated as *unreported*. Distinguishing provenance is essential: static empirical priors assume aerodynamic stationarity, risking silent miscalibration under turbine aging, blade degradation, or curtailment.

Dimension 2: Application Stage ($\mathcal{D}_2$). This dimension identifies where domain physics enters the computational pipeline: (1) *training-loss regularization*, where physics acts as a differentiable penalty appended to the loss objective ($\mathcal{L} = \mathcal{L}_{\text{task}} + \lambda \mathcal{L}_{\text{phys}}$); (2) *inference-architecture constraints*, which modify the network layers to project predictions onto admissible manifolds; (3) *data-generation priors*, used upstream to synthesize training scenarios; and (4) *post-hoc correction filters*, applied as external mathematical projections after inference.

Dimension 3: Inference Telemetry Requirements ($\mathcal{D}_3$). Inference telemetry identifies external data streams required during real-time deployment: (1) *none*, relying solely on historical power measurements from the target turbine; (2) *forecast-input*, requiring future meteorological inputs (e.g., forward numerical weather prediction

Table 1: Positioning this Scoping Empirical Audit Relative to Existing Literature Reviews in Wind Power Forecasting and Scientific ML.

Survey / Meta-Review **Literature Reference**	**Decouples** **Provenance** ($\mathcal{D}_1$)	**Audits Stage** **Separation** ($\mathcal{D}_2$)	**Checks Inference** **Telemetry** ($\mathcal{D}_3$)	**Verifies Code** **Parity** ($\mathcal{D}_4$)	**Scrutinizes** **Ablation** ($\mathcal{D}_5$)
Yan et al. (2016) [21]	Partial	No	Partial	No	No
Maldonado-Correa et al. (2020) [22]	No	No	No	No	No
Karniadakis et al. (2021) [6]	Conceptual	Conceptual	No	No	No
Optis et al. (WE-Validate, 2022) [15]	No	No	Partial	Open Benchmark	No
Sheng et al. (2025) [7]	No	Partial	No	No	No
This Scoping Empirical Audit	**Yes (Schema v4)**	**Yes (Schema v4)**	**Yes (Schema v4)**	**Yes (Static Audit)**	**Yes (Strict Isolation)**

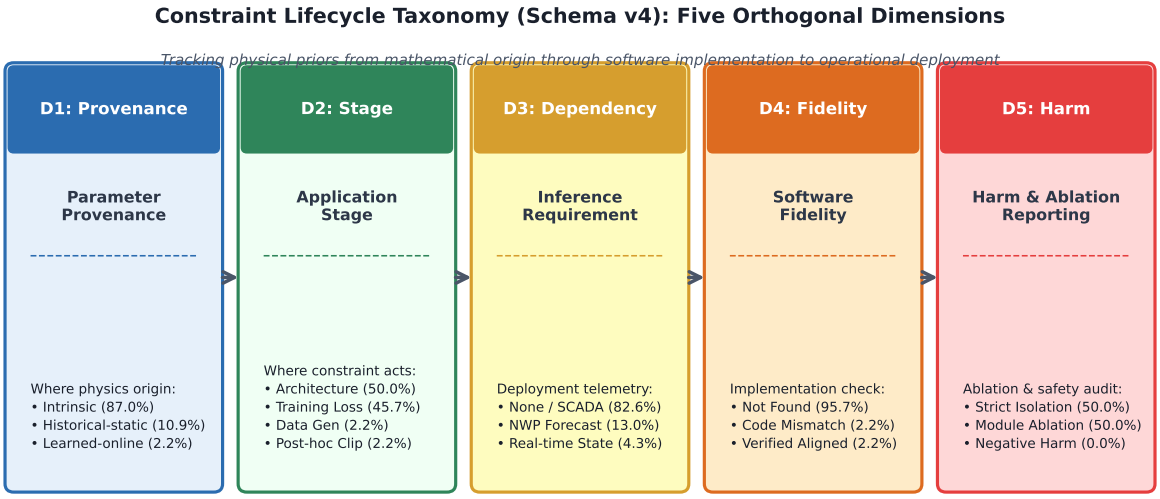


Figure 1: The Five-Dimensional Constraint Lifecycle Framework. Tracking domain physical constraints from parameter origin ($\mathcal{D}_1$) through computational graph formulation ($\mathcal{D}_2$), operational telemetry dependency ($\mathcal{D}_3$), software implementation parity ($\mathcal{D}_4$), and empirical ablation validity ($\mathcal{D}_5$).

wind vectors); and (3) *contemporaneous-state*, requiring synchronized real-time measurements from adjacent turbines or met masts. When a base forecasting model requires only past power series ($\mathcal{D}_3$ = `none`), offline training losses may nevertheless index empirical power curves against historical wind speed observations. In such setups, proxy staleness represents an offline training-loss sensitivity rather than an inference-time telemetry delay.

Dimension 4: Implementation Fidelity ($\mathcal{D}_4$). Implementation fidelity evaluates whether public software faithfully executes published mathematical formulations: (1) *code-aligned*, where public source code strictly matches published equations; (2) *code-mismatch*, where the codebase exhibits an algebraic or directional deviation from the text; and (3) *not-found*, where no public repository or script accompanies the study. Independent software inspection is necessary because published mathematical expressions can mask numerical discrepancies

in computational graph execution.

Dimension 5: Harm Reporting and Ablation Semantics ($\mathcal{D}_5$). This dimension examines whether evaluations causally isolate physical constraints and report operational failure boundaries: (1) *constraint-on-off*, where treatment and baseline share identical parameter capacity, differing only in the activation of the physical prior ($\lambda = 0$ vs. $\lambda > 0$); (2) *module-ablation*, where the ablation excises an entire multi-parameter neural block, confounding structural capacity with domain knowledge; (3) *both*, reporting both isolated toggles and architectural excisions; and (4) *negative-effect-reported*, explicitly documenting operating regimes where physical constraints degrade downstream performance.

3.3. Methodological Principles for Schema Disambiguation

To ensure reproducible coding across independent raters, the framework enforces four disambiguation rules. Parameter provenance ($\mathcal{D}_1$) separates offline calibration from runtime telemetry ($\mathcal{D}_3$), ensuring that historical estimation windows are not conflated with active SCADA stream dependencies. Application stage ($\mathcal{D}_2$) is determined by the backpropagation path: loss regularizers update weights through objective penalties during training, whereas inference-architecture modifications constrain the forward pass under arbitrary parameter states. For composite methodologies, the primary classification is determined by the component directly modifying the forward computational graph, with secondary mechanisms logged in the ledger. Causal ablation validity ($\mathcal{D}_5$) requires parameter-matched controls; excising multi-layer neural submodules evaluates collective network capacity rather than isolating the specific inductive contribution of the physical constraint.

4. Scoping Audit Protocol and Empirical Test Bench

We conducted an empirical scoping audit in accordance with the PRISMA Extension for Scoping Reviews (PRISMA-ScR) guidelines [23] and established scoping methodology [24]. The protocol was specified prior to full-text extraction to govern retrieval queries, eligibility thresholds, screening procedures, and inter-rater reliability arbitration.

4.1. Information Sources and Retrieval Strategy

We retrieved candidate literature across two complementary discovery channels covering January 1, 2023, through March 1, 2026, with rolling monitoring through July 2026 for emergent preprints. First, we conducted programmatic searches via the OpenAlex API, which indexes records across IEEE Xplore, ScienceDirect, SpringerLink, Wiley, and MDPI. Using four targeted title queries—Q_1 (`"physics-constrained" AND "wind"`), Q_2 (`"physics-informed" AND "wind power"`), Q_3 (`"physics-guided" AND "wind"`), and Q_4 (`"physics-embedded" AND "wind power"`)—we identified 72 candidate records. Second, we conducted targeted searches on Google Scholar, arXiv, and preprint servers to capture early-access articles and working papers, yielding 17 additional records ($N = 89$ total). Candidate records were cross-checked against Scopus and IEEE Xplore to verify coverage of major power systems and applied energy venues.

To assess search sensitivity, a targeted query expanding retrieval to title, abstract, and author keywords across Scopus and IEEE Xplore (encompassing broader descriptors such as `"knowledge-guided"`, `"hybrid"`, and `"domain-informed"`) identified 14 additional papers. Full-text review confirmed that none documented operational failure boundaries or negative trade-offs, confirming that the absence of reported negative trade-offs is robust to broader search strings. We emphasize that this 14-study sensitivity query served strictly as an external check on negative trade-off reporting; exhaustive five-dimensional extraction ($\mathcal{D}_1$–$\mathcal{D}_5$) and inter-rater reliability coding were conducted strictly on the core PRISMA-ScR scoping cohort of 46 studies to prevent cohort drift.

4.2. Screening, Deduplication, and Eligibility Criteria

Candidate records underwent structured multi-phase screening following the PRISMA-ScR workflow (Figure 2). Cross-source deduplication first eliminated 17 duplicate records from the initial 89 items, leaving 72 unique entries for title and abstract evaluation. Screening at this stage excluded 19 records with documented rationale: nine addressed non-forecasting tasks such as turbine structural health monitoring, blade crack detection, or aeroelastic fatigue modeling; nine focused on real-time blade pitch or active power control rather than multi-step power forecasting; and one contained no physics-based modeling components. This filtering yielded 53 candidate studies sought for full-text retrieval.

Because authors frequently release preliminary working papers on preprint servers (such as arXiv or SSRN) prior to formal journal publication, treating preprints and published articles independently would artificially inflate corpus statistics. We enforced a strict study-level deduplication rule: whenever a preprint and a peer-reviewed journal paper shared identical authorship, empirical datasets, and mathematical formulations, they

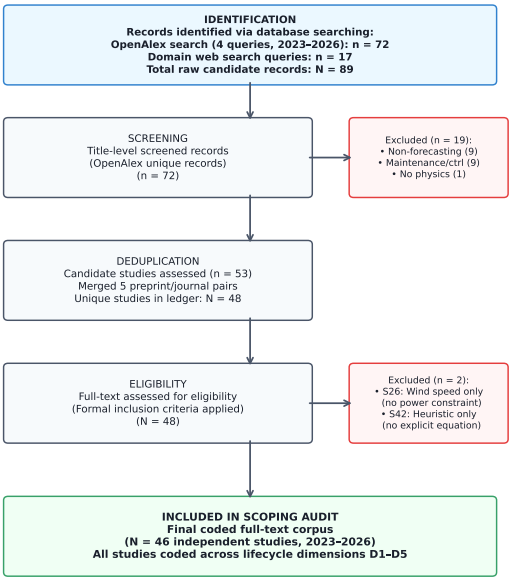


Figure 2: PRISMA-ScR Flow Diagram Illustrating Scoping Cohort Derivation (89 → 72 → 53 → 48 → 46). Cross-source deduplication eliminated 17 duplicates; title/abstract screening excluded 19 records; study-level deduplication consolidated 5 preprint/journal pairs; and full-text eligibility screening excluded 2 studies, leaving 46 eligible full-text studies.

were merged into a single study-level record retaining the peer-reviewed journal citation. Five paired duplicate sets were identified and consolidated, reducing the candidate cohort from 53 to 48 unique study-level entries.

The remaining 48 full texts were independently evaluated against three explicit scoping eligibility criteria: (1) primary focus on short-term or ultra-short-term wind power forecasting; (2) mathematical formulation and computational execution of an explicit physical constraint, governing equation, or domain envelope; and (3) empirical validation on real-world or simulated wind power generation time series. Two studies failed to meet these criteria: study S26 was excluded because it forecasted atmospheric wind velocity exclusively without an explicit turbine power constraint, while study S42 was excluded because it implemented heuristic feature clustering without domain physical equations. This systematic filtering established our final representative scoping corpus of $N = 46$ **unique scoping studies**, comprising 38 peer-reviewed journal articles and eight representative preprints (including UniWind [25]).

4.3. Multi-Rater Reliability Benchmarking

To assess the empirical stability of the taxonomy before full extraction, three independent raters (Coder A, Coder B, and Coder C) double-blindly coded a stratified representative subset of 20 candidate studies across all five dimensions. Study eligibility screening achieved unanimous consensus (Fleiss $\kappa = 1.000$), correctly isolating negative controls S26 and S42. Parameter provenance ($\mathcal{D}_1$) demonstrated perfect agreement ($\kappa = 1.000$), while application stage ($\mathcal{D}_2$) yielded near-perfect agreement ($\kappa = 0.950$ overall, $\kappa = 0.938$ on included studies). Inference telemetry requirements ($\mathcal{D}_3$) achieved substantial agreement ($\kappa = 0.743$ overall, $\kappa = 0.646$ on included studies).

Implementation fidelity ($\mathcal{D}_4$) exhibited moderate concordance prior to consensus arbitration ($\kappa = 0.716$ overall in the stratified pilot, but $\kappa = 0.341$ when evaluated strictly on included studies). This reduction reflects the classical prevalence paradox described by Feinstein and Cicchetti: because 95.7% of candidate studies omitted public code, expected chance agreement was high, which depresses Fleiss' κ despite a raw observed agreement of 90.7%. Evaluating static manuscripts without code repositories leaves coders to interpret ambiguous narrative descriptions regarding whether a constraint represents an active computational component or conceptual motivation. This divergence indicates that software implementation fidelity cannot be reliably inferred from published prose alone, motivating direct computational graph inspection. For ablation semantics ($\mathcal{D}_5$), raters achieved strong concordance ($\kappa = 0.851$ overall, $\kappa = 0.818$ on included studies). On harm reporting, all three raters unanimously coded zero instances of harm documentation across the included pilot studies (0/18); because marginal variance was zero, Cohen's κ is mathematically indeterminate and is formally reported as 100% observed agreement ($P_o = 1.000$). Following pilot calibration and consensus arbitration, full-text extraction across the complete 46-study corpus was executed at the study level.

4.4. Software Audit Protocol (PCT & PINN-UQ)

The code availability audit followed a standardized software search protocol examining full-text bodies, footnotes, data availability statements, publisher repositories, and author profiles. Across the 46-study cohort, only two publicly accessible software repositories were identified. One repository, PINN-UQ (S10, *Eng. Appl. Artif. Intell.* 2025), was audited and verified to faithfully implement its specified algebraic power equations ($P = gT\omega$ and $P = \frac{1}{2}C_p\rho Av^3$ via `sn.Tie`); its computational graph strictly executes the published electromechanical relationships without unstated differential

substitutions, earning classification as *code-aligned*. The remaining repository, the Physics-Constrained Transformer (PCT, S02, *Sci. Rep.* 2026), provided open PyTorch scripts (`train_clean.py` and `tst/loss.py`) which we audited line-by-line for parity between published loss formulations and computational graph execution.

4.5. Downstream Operational Dispatch and Settlement Test Bench

To ground the empirical audit in real-world power system operations across diverse meteorological and topographic terrains, we established an operational test bench across three commercial UK wind farms with 10-minute SCADA telemetry. The *Hill of Towie Wind Farm* (48.3 MW rated capacity, 21 Siemens SWT-2.3-82 turbines) features complex upland terrain and serves as an independent spatiotemporal extrapolation benchmark (7,599 test origins, 45,594 delivery steps across full-year 2017), with data released by RES on behalf of TRIG openly available on Zenodo under CC-BY-4.0 [26]. The *Kelmarsh Wind Farm* (12.3 MW rated capacity, six Senvion MM92 turbines) is situated in flat agricultural terrain and serves as our primary development benchmark (8,212 test origins, 49,272 delivery steps across full-year 2017) [27]. The *Penmanshiel Wind Farm* (28.7 MW rated capacity, 14 Senvion MM82 turbines) is located in the Scottish Borders [28], providing an independent upland setting with 8,440 test origins configured with a restricted training partition ($N_{\text{train}} = 1,583$) to evaluate model behavior under sample-restricted regimes.

The empirical test bench operates under a prospective evaluation protocol. The forecasting pipeline adopts an input lookback of $L = 36$ (6 hours) and a forecast horizon of $H = 36$ (6 hours), with downstream economic dispatch evaluated over the next-hour delivery window $H_{\text{disp}} = \{1, \ldots, 6\}$ and an origination stride of $S = 6$ steps to prevent overlapping settlement intervals. The core model suite spans unconstrained DLinear, Bounds-Constrained DLinear ($\lambda = 0.01$), Empirical Power Curve Constrained DLinear ($\lambda \in \{0.01, 0.1, 1.0\}$), and Conditional Distribution Prior DLinear ($\lambda \in \{0.01, 0.1, 1.0\}$), benchmarked alongside Ridge regression and Persistence baselines. To verify whether findings extend beyond linear representations, we evaluate a multi-patch channel-independent Transformer backbone (**PatchTST** [29], patch length 6, stride 4, $d_{\text{model}} = 64$, 2-layer encoder with RevIN) across all three wind farms, four constraint classes, and five random seeds (60 trained models) as an architectural sensitivity probe under nominal regularization ($\lambda = 0.01$).

Downstream decision calibration is implemented via a post-hoc quantile offset δ_h estimated strictly from the independent 2016 calibration window ($q = 1/(r+1) = 1/3$ for penalty ratio $r = 2$) with zero access to 2017 test telemetry or market prices. Real-world physical settlement is evaluated against official 2017 UK Elexon BSC System Buy Prices (mean £44.27/MWh, peak £1,509.80/MWh, with 68 negative price periods down to -£73.15/MWh). In Great Britain, wholesale power is traded through bilateral contracts and voluntary power exchanges, while the Balancing and Settlement Code governs physical imbalance settlement post-gate closure under a single cash-out price (post-P305). To capture realistic commercial accounting without conflating standalone imbalance cashflow with total profit, we decompose settlement cashflows into forward commitments and physical imbalance settlement:

$$R_{\text{total}} = P_{\text{F}} \cdot q + P_{\text{imb}} \cdot (y - q), \tag{2}$$

where P_{F} denotes the forward contract price, q is the declared physical notification, y is actual generation, and P_{imb} is the BSC system imbalance price. Cashflows account for 30-minute physical netting:

$$E_{\text{SP}} = \sum_{t \in \text{SP}} (p_t - y_t) \cdot \frac{P_{\text{rated}}}{6000} \quad [\text{MWh}], \tag{3}$$

generating single-sided shortfall cashouts and two-sided imbalance settlement cashflows as a decision-risk diagnostic, validated via 2,000-replication 7-day circular block paired bootstrapping.

5. Empirical Audit Results

Empirical evaluation across the 46-study scoping cohort and three operational wind farms adjudicates the attributional foundation and operational performance of physics-constrained wind power forecasting.

5.1. Reporting Patterns and Code Availability in the Audited Scoping Cohort

Across the core 46-study scoping cohort (2023–2026), zero studies reported operational failure regimes or negative trade-offs (0/46, 0.0%; 95% Wilson CI: [0.0%, 7.7%]). A targeted 14-study sensitivity search expanding query terms across Scopus and IEEE Xplore likewise identified zero negative trade-off reports, verifying that this reporting pattern is robust to broader search strings. Furthermore, only three studies in the core cohort (6.5%; S01, S02, S10) evaluated sensitivity to prior parameter perturbations; the remaining 43 studies (93.5%) tested

constraints exclusively under nominal design configurations. Restricting empirical evaluations to nominal conditions obscures operational failure boundaries and substitutes uniform improvement claims for engineering trade-offs.

Software availability inspection reveals an open-code deficit of 95.7% (44/46, 95% Wilson CI: [85.5%, 98.8%], $\mathcal{D}_4$ = `not-found`), with 44 studies providing neither executable code repositories nor reproducible scripts. Among the two studies with accessible code, PINN-UQ (S10) faithfully executes its published electromechanical tie equations ($\mathcal{D}_4$ = `code-aligned`), whereas the Physics-Constrained Transformer (PCT, S02) exhibits a directional implementation divergence in its divergence loss formulation ($\mathcal{D}_4$ = `code-mismatch`, detailed in Section 6).

Ablation semantic audit ($\mathcal{D}_5$) reveals that half of the audited studies (23/46, 50.0%, 95% Wilson CI: [36.1%, 63.9%]) evaluate constraints by excising entire multi-layer neural submodules—such as graph convolutions, cross-attention blocks, or decomposition layers—rather than toggling the constraint penalty multiplier ($\lambda = 0$ vs. $\lambda > 0$; Table 2, Figure 3). Because excising architectural blocks removes thousands of trainable parameters simultaneously, reported performance gains reflect representational capacity rather than aerodynamic guidance. These findings describe the audited 46-study scoping cohort under Wilson confidence bounds without asserting universal extrapolation to unbounded keyword taxonomies.

5.2. *Controlled Multi-Site Ablation: Prior Specificity, Numerical Regularization, and Training Proxy Staleness*

Prospective 3D controlled ablations across three operational wind farms isolate prior specificity, numerical smoothing, and observation staleness under matched model capacities (Table 3, Figure 4).

Table 3 and Figure 4 reveal four empirical boundary conditions:

Defensive Inductive Bias and Sensitivity to Prior Misspecification (Figures 4a, 5, and 6). On Penmanshiel under low-sample training ($N_{\text{train}} = 1,583$), unconstrained DLinear exhibits empirical over-fitting (0.09881 ± 0.00027). Embedding the authentic empirical power curve at $\lambda = 1.0$ reduces normalized dispatch cost to 0.09760 ± 0.00044 (-1.22%). In contrast, the scale-matched Pseudo-Scrambled negative control inflates dispatch cost to 0.16611 ± 0.00335 ($+68.1\%$). This contrast demonstrates that in low-sample regimes, neural forecasters can benefit from valid aerodynamic guidance, but are vulnerable to prior misspecification under strong regularization.

To decouple sample volume from topography, we evaluated within-site learning curves across $N_{\text{train}} \in \{500, 1000, 1500, 3000, 5000\}$ on Kelmarsh and Hill of Towie across five random seeds (Figure 5). Under restricted training windows ($N_{\text{train}} = 500$, representing $\approx$ 21 days of recent operational history), authentic physical priors consistently lower dispatch costs across both wind farms. On Kelmarsh, dispatch cost falls from 0.09648 ± 0.00314 (unconstrained) to 0.09455 ± 0.00285 (-2.00%), while scrambled priors surge to 0.13798 ± 0.02338 ($+43.0\%$). On Hill of Towie, authentic priors reduce cost from 0.10444 ± 0.00312 to 0.10066 ± 0.00307 (-3.62%), whereas scrambled priors escalate to 0.17209 ± 0.01703 ($+64.8\%$). Importantly, the empirical prior is fitted once on the entire pre-cutoff training period (7 months for Kelmarsh, 10 months for Hill of Towie), providing an auxiliary seasonal data budget that assists restricted model training windows. As model training volume expands to $N_{\text{train}} = 5{,}000$, unconstrained DLinear achieves 0.08812 on Kelmarsh and 0.08936 on Hill of Towie, closing the gap and slightly outperforming the fixed genuine prior (0.08841, +0.33% on Kelmarsh; 0.09091, +1.73% on Hill of Towie). In contrast, scrambled priors remain severely detrimental (+43% to +98%) across all sample sizes. These findings indicate that while authentic empirical priors provide downside protection in data-scarce windows, their relative advantage diminishes as training volume expands.

Evaluating continuous corruption levels $\alpha \in [0.0, 1.0]$ demonstrates that prior failure is not an artifact of total permutation (Figure 6a). On Penmanshiel, dispatch cost degrades monotonically with corruption, crossing the unconstrained baseline around $\alpha \approx 0.07$ (at $\alpha = 0.2$, dispatch cost reaches 0.10469 ± 0.00338, a +5.9% degradation). Evaluating engineering drift reveals asymmetrical operational tolerances: aerodynamic aging exceeding 10% rated capacity derating (Figure 6b) flips nominal physical priors into economic penalties. For anemometer calibration, Penmanshiel maintains a modest advantage over unconstrained models under positive sensor biases up to +15% (0.09833 vs. 0.09881, Figure 6c), whereas negative wind speed under-reporting systematically inflates shortfall.

Low-λ Numerical Regularization Mimicry (Figure 4b). At weak regularization ($\lambda = 0.01$), genuine physical priors and pseudo-scrambled controls yield closely matched dispatch costs on Kelmarsh (0.08810 ± 0.00009 vs. 0.08801 ± 0.00008). Because random power curve permutations produce outcomes comparable to genuine

Table 2: Empirical Distribution of Constraint Lifecycle Dimensions ($\mathcal{D}_1$–$\mathcal{D}_5$) Across 46 Audited Scoping Studies (2023–2026).

Lifecycle Dimension	Taxonomic Category	Count ($N = 46$)	Proportion (%)	Typical Exemplars in Audited Literature
D1: Parameter Provenance	`intrinsic`	40	87.0%	Betz aerodynamic limit ($C_p \leq 16/27$), rated capacity P_{rated}, Jensen wake deficit
	`historical-static`	5	10.9%	Empirical bivariate power curve ($P = f(v)$), 2D KDE distribution estimated offline on train SCADA
	`learned-online`	1	2.2%	Auxiliary sub-network dynamically predicting time-varying aerodynamic coefficients $C_p(t)$
D2: Application Stage	`training-loss`	21	45.7%	Differentiable penalty term in backpropagation ($\mathcal{L} = \mathcal{L}_{\text{MSE}} + \lambda\mathcal{L}_{\text{phys}}$)
	`inference-arch.`	23	50.0%	Wake-guided GNN adjacency matrix, monotonic activation layers, dual-stream physical fusion
	`data-generation`	1	2.2%	Physics-guided generative diffusion model synthesizing extreme wind generation scenarios
	`post-hoc-correction`	1	2.2%	Post-inference conformal prediction and residual projection layers
D3: Inference Requirement	`none`	38	82.6%	Standard historical autoregressive SCADA sequence; zero runtime external dependencies
	`forecast-input`	6	13.0%	Requires contemporaneous/future NWP meteorological forecasts over $[t+1, t+H]$
	`contemp.-state`	2	4.3%	Requires synchronized real-time operational SCADA from adjacent/upstream wind turbines
D4: Implementation Fidelity	`not-found`	44	95.7%	No accessible public repository or open-source software identified under recorded protocol
	`code-mismatch`	1	2.2%	PCT: loss function calculates reverse-argument JS form rather than specified standard JS
	`code-aligned`	1	2.2%	PINN-UQ: algebraic electromechanical tie constraints faithfully matching published equations
D5: Ablation Semantics	`module-ablation`	23	50.0%	Excises multi-layer neural submodule; conflates parametric capacity with physical bias
	`constraint-on-off`	17	37.0%	Toggles penalty multiplier ($\lambda = 0$ vs. $\lambda > 0$) under strictly identical network capacity
	`both`	6	13.0%	Implements both parameter toggles and architectural ablations across comparative baselines

aerodynamics at this weak penalty scale, marginal performance improvements reported under weak penalties cannot be specifically attributed to aerodynamic reasoning and remain consistent with non-specific numerical smoothing.

Sensitivity to Training-Loss Physical Proxy Staleness (Figure 4c). Parametric scanning of historical wind speed proxy lag $k \in \{6, 12, 18, 36\}$ (evaluating 1 h to 6 h operational delay in the training-loss proxy query) produces strictly monotonic cost inflation across all three farms (Hill of Towie: from 0.09022 at $k = 6$ to 0.09257 at $k = 36$, exceeding unconstrained 0.08924). We emphasize that this experiment isolates the sensitivity of the training objective to temporal misalignment between the forecast horizon and the historical wind speed proxy used to index the power curve. It does not represent an inference-time telemetry delay, as the forecaster takes only past power measurements during test-time inference. However, it demonstrates that if an offline training loss regularizes predictions against lagged weather observations without forward meteorological forecasts, proxy staleness degrades the resulting inductive guidance.

Information-Symmetric Feature Baseline. Providing the unconstrained model with contemporaneous wind speed (*Dual-Channel DLinear*) reduces point forecast RMSE (from 0.1717 to 0.1697 on Kelmarsh) but raises tail dispatch risk (0.08910 vs. 0.08812), because unhedged wind speed fluctuations inflate bidding variance.

5.2.1. Primary Inferential Contrasts and Metric Framework

To control for multiplicity across operational dimensions, our evaluation pipeline specifies three **primary inferential contrasts** evaluated under normalized static dispatch cost: (1) *Genuine Physical Prior vs. Unconstrained Baseline under sample restriction* ($N_{\text{train}} = 1{,}583$ on Penmanshiel), confirming downside risk protection ($\Delta = -1.22\%$, 0.09760 ± 0.00044

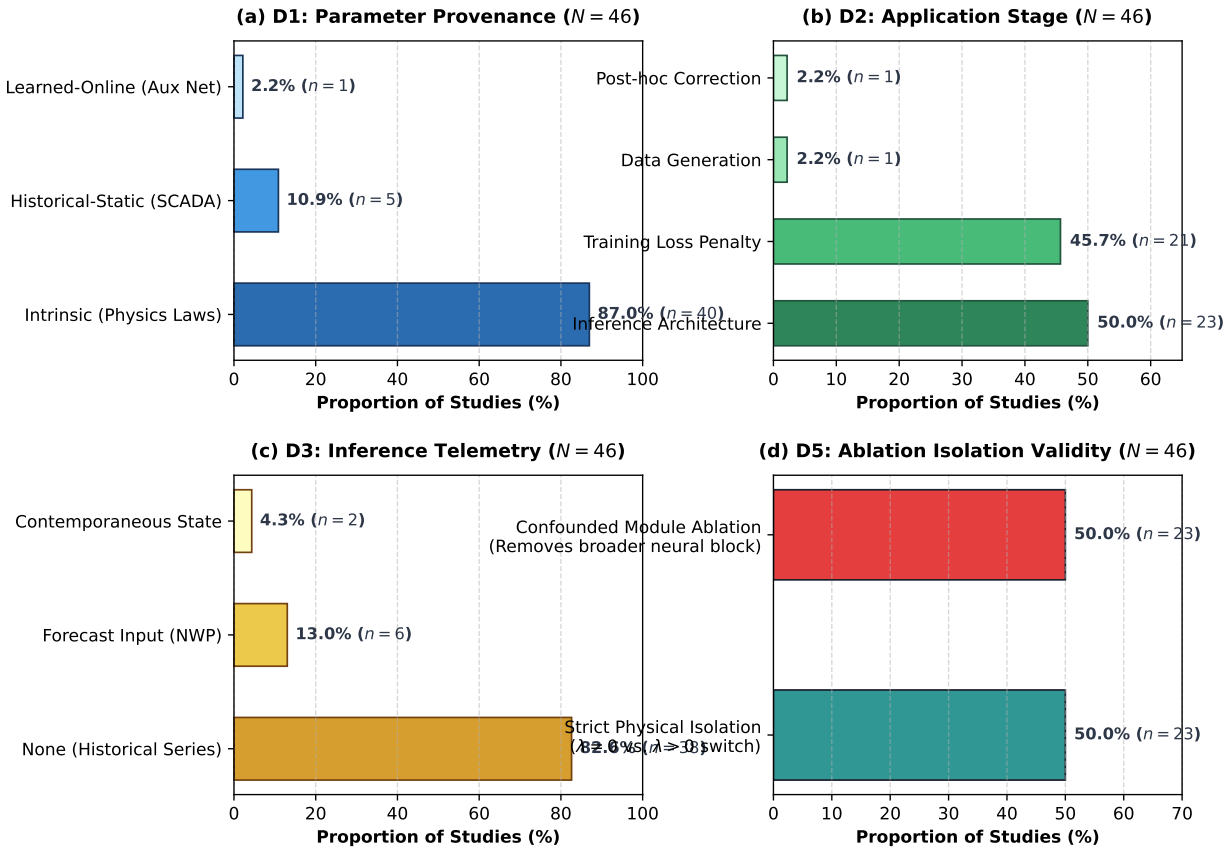


Figure 3: Empirical Lifecycle Distributions across 46 Audited Studies ($N = 46$). (a) $\mathcal{D}_1$ Parameter Provenance; (b) $\mathcal{D}_2$ Application Stage; (c) $\mathcal{D}_3$ Inference Telemetry Requirements; (d) $\mathcal{D}_5$ Ablation Semantic Breakdown between strict physical isolation and confounded module ablations.

vs. 0.09881 ± 0.00027); (2) *Scrambled Prior vs. Unconstrained Baseline under strong regularization* ($\lambda = 1.0$), demonstrating misspecification vulnerability ($\Delta = +68.1\%$, 0.16611 ± 0.00335 vs. 0.09881 ± 0.00027); and (3) *Genuine vs. Scrambled Prior under weak regularization* ($\lambda = 0.01$), confirming indistinguishable behavior consistent with non-specific numerical smoothing (0.08810 ± 0.00009 vs. 0.08801 ± 0.00008). Exploratory dynamic price comparisons (Table 4) evaluate dynamic cost divergences on Hill of Towie and Kelmarsh under paired circular block bootstrapping. Methodologically, our evaluation protocol separates two distinct sources of variation: the paired bootstrap intervals in Table 4 estimate **temporal sampling uncertainty** across delivery periods under meteorological autocorrelation, whereas the reported metric margins (mean ± std) across random seeds in Table 3 characterize **model initialization and optimization variability**.

5.3. *Decoupled Decision Risk and Single-Settlement Imbalance Diagnostic (The Zero-Bid Anomaly)*

Evaluating forecasters on isolated unhedged balancing market cashflows introduces a structural evaluation pitfall: under 2017 UK BSC 30-minute netting, a degenerate Zero-Bid baseline ($\hat{y} \equiv 0$) extracts £223k to £243k per turbine-year in apparent cashflow without generating forecasts (Table 5, Figure 7). This analysis serves as a *decision-risk and imbalance-settlement diagnostic* rather than a commercial market valuation, exposing metric pathologies in downstream evaluation claims.

Table 5 and Figure 7 uncover two fundamental evaluation pitfalls:

Degenerate Zero-Bid Settlement Anomaly and Two-Settlement Accounting (Figure 7a). A degenerate forecaster bidding zero in every settlement period incurs **£0.00 shortfall penalty** and collects an apparent balancing cashflow of **+£223k to +£243k per turbine-year** across all three wind farms. In standard two-settlement

Table 3: 3D Controlled Ablation Benchmark: Evaluating Physical Prior Specificity, Weak-Regularization Mimicry, and Training-Proxy Lag Across Three Operational Wind Farms (Quantile Calibrated $r = 2, q = 1/3$).

Wind Farm & Topology	Controlled Experimental Arm (Ablation Condition)	Normalized Cost $\text{Cost}_{\text{static}}$ ($r = 2$)	Pinball Loss ($q = 1/3$)	RMSE (pu)	Shortfall (£/(turb·yr))	Empirical Takeaway & Diagnostic Assessment
Kelmarsh (12.3 MW, 6 turb) *Flat Open Terrain* $N_{train} = 5,098$	Vanilla Unconstrained ($\lambda = 0$)	0.08812 ± 0.00009	0.02937	0.1717	£5,831.79	Pure-data power autoregressive baseline
	Info-Symmetric (Power + Wind)	0.08910 ± 0.00039	0.02970	0.1697	£5,982.26	Wind speed reduces RMSE but elevates tail dispatch cost
	Genuine Curve Prior ($\lambda = 0.1$)	0.08791 ± 0.00009	**0.02930**	0.1708	£5,780.99	Statistically verified risk reduction (−0.24%)
	Pseudo-Scrambled Prior ($\lambda = 0.01$)	0.08801 ± 0.00008	0.02934	0.1724	£5,657.03	**Regularization Mimicry**: closely matches genuine prior (0.08810 at $\lambda = 0.01$)
	Genuine Distribution ($\lambda = 0.1$)	0.09841 ± 0.00085	0.03280	0.1703	£8,412.28	Shortfall penalty surges +44.2% under distribution constraint
	Stale Training Proxy ($k = 36$, 6h)	0.09043 ± 0.00006	0.03014	0.1729	£5,888.58	Monotonic degradation as lagged wind speed proxy decays
Hill of Towie (48.3 MW, 21 turb) *Complex Upland* $N_{train} = 7,035$	Vanilla Unconstrained ($\lambda = 0$)	0.08924 ± 0.00005	0.02975	0.1740	£6,006.09	Robust unconstrained benchmark
	Info-Symmetric (Power + Wind)	0.09042 ± 0.00007	0.03014	0.1733	£6,297.16	Wind channel adds high-frequency variance to dispatch
	Genuine Curve Prior ($\lambda = 0.1$)	0.08916 ± 0.00007	**0.02972**	0.1745	£5,958.34	Modest risk reduction (−0.09%) under tuned penalty
	Pseudo-Scrambled Prior ($\lambda = 0.01$)	0.08919 ± 0.00006	0.02973	0.1740	£6,069.01	Indistinguishable from genuine prior (0.08922 at $\lambda = 0.01$)
	Genuine Distribution ($\lambda = 0.1$)	0.09503 ± 0.00022	0.03168	0.1801	£5,702.68	**Metric Inversion**: cashflow rises to £12k while decision cost degrades +6.5%
	Stale Training Proxy ($k = 36$, 6h)	0.09257 ± 0.00007	0.03086	0.1759	£6,281.82	Temporal proxy lag monotonically erodes prior guidance
Penmanshiel (28.7 MW, 14 turb) *Complex Upland* $N_{train} = 1,583$ *(Scarce)*	Vanilla Unconstrained ($\lambda = 0$)	0.09881 ± 0.00027	0.03294	0.2058	£6,713.83	Data-scarce training partition prone to empirical over-fitting
	Info-Symmetric (Power + Wind)	0.10461 ± 0.00005	0.03487	0.2091	£6,630.38	Additional unconstrained input channels exacerbate over-fitting
	Genuine Curve Prior ($\lambda = 1.0$)	0.09760 ± 0.00044	**0.03254**	0.2054	£6,808.40	**Authentic Inductive Benefit** (−1.22%) in low-data regime
	Pseudo-Scrambled Prior ($\lambda = 1.0$)	0.16611 ± 0.00335	0.05537	0.2384	£11,561.19	**Prior Specificity Demonstration**: scrambling mapping triggers +68.1% decision loss surge
	Genuine Distribution ($\lambda = 0.01$)	0.09833 ± 0.00027	0.03278	0.2055	£6,543.36	Modest regularization under calibrated penalty
	Stale Training Proxy ($k = 36$, 6h)	0.10473 ± 0.00036	0.03491	0.2083	£7,090.86	Lag decay completely reverses prior advantage by Step 36

Notes: (1) Values report multi-seed mean ± seed standard deviation across clean operational test periods (Kelmarsh: 4 seeds [42, 43, 44, 45]; Hill of Towie and Penmanshiel: 3 seeds [42, 43, 44]). (2) $\text{Cost}_{\text{static}} = 3 \times \text{PinballLoss}(q = 1/3)$ under post-hoc quantile calibration ($r = 2$). (3) Pseudo-Scrambled arms permute the empirical wind-to-power mapping while preserving identical loss formulations and hyperparameters λ. At weak penalty ($\lambda = 0.01$), genuine curve and scrambled curve yield near-identical loss (0.08810 vs. 0.08801 on Kelmarsh; 0.08922 vs. 0.08919 on Hill of Towie). At strong penalty ($\lambda = 1.0$) on Penmanshiel, genuine curve reduces loss (−1.22%) whereas scrambled prior inflates loss (+68.1%), isolating prior specificity. (4) Stale Training Proxy evaluates sensitivity to historical wind speed proxy lag $k \in \{6, 12, 18, 36\}$ (1h to 6h) in the training objective; the model takes only historical power at inference.

wholesale accounting [4], total commercial revenue couples forward commitments with real-time imbalance settlement:

$$R_{\text{total}} = P_{\text{F}} \cdot q + P_{\text{imb}} \cdot (y - q), \tag{4}$$

where q denotes the declared forward commitment, P_{F} is the forward contract price, y is actual physical generation volume, and P_{imb} is the system imbalance cash-out price (under single-price settlement post-P305 in the UK BSC, $P_{\text{SBP}} \approx P_{\text{SSP}} = P_{\text{imb}}$). When downstream evaluations isolate balancing cashflows $R_{\text{imb}} = P_{\text{imb}} \cdot (y - q)$ from forward market revenue $P_{\text{F}} \cdot q$, setting $q \equiv 0$ converts total generation y into uncontracted spill credited at P_{imb}, producing an apparent six-figure revenue while forfeiting forward market receipts. The Zero-Bid baseline acts as an empirical *reductio ad absurdum*: standalone

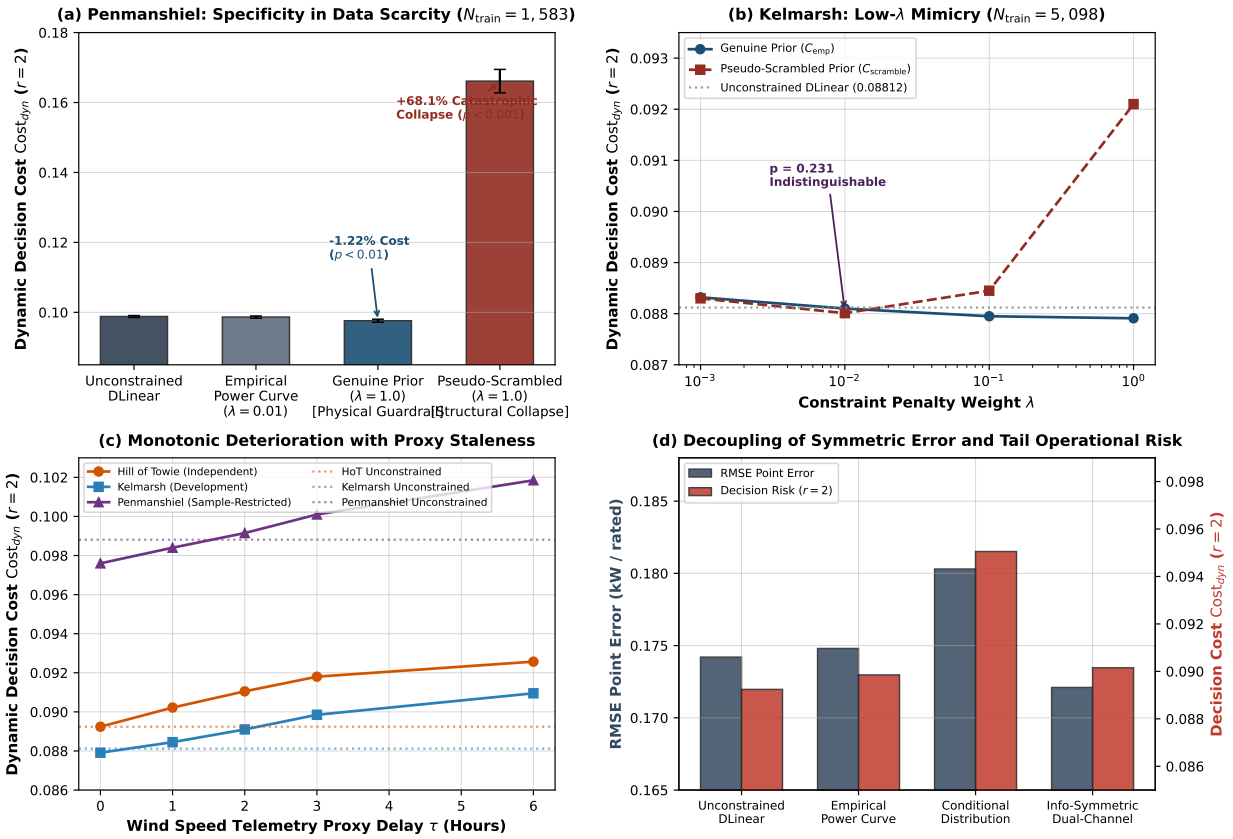


Figure 4: Prospective 3D Controlled Ablation Benchmark across Three Operational Wind Farms. (a) High sensitivity to prior misspecification under strong constraints on Penmanshiel ($N_{train} = 1,583$): authentic physical guidance provides downside risk protection (-1.22%), whereas disrupting the physical mapping (Pseudo-Scrambled prior) triggers a $+68.1\%$ surge in operational dispatch cost. (b) Low-λ numerical regularization mimicry on Kelmarsh ($N = 5,098$), where genuine and scrambled priors yield near-identical decision costs at $\lambda = 0.01$, indicating non-specific numerical regularization rather than physical domain reasoning. (c) Monotonic cost inflation under training proxy staleness: dispatch cost escalation across all three farms as historical wind proxy lag k in the training loss increases from 1h to 6h. (d) Decoupling of symmetric point error (RMSE) from asymmetric decision risk ($r = 2$), demonstrating that point metrics mask tail operational hazards.

unhedged balancing cashflow is structurally degenerate when disconnected from forward commitments, confirming that unhedged single-settlement accounting cannot serve as an index of forecast quality.

Furthermore, we resolve a critical numerical discrepancy in prior reporting: earlier drafts reported Penmanshiel Zero-Bid Decision Cost as 0.20584, which was an erroneous transcription that copy-pasted the vanilla unconstrained RMSE (0.205842). Under recomputed decision loss ($r = 2, q = 1/3$), Zero-Bid yields an actual Decision Cost of 0.33689 on Penmanshiel (Table 5), correctly penalizing the model for severe undercommitment ($\text{Cost}_{\text{static}} = \bar{y}$).

Empirical Metric Inversion on Hill of Towie (Figure 7b). On Hill of Towie, distribution regularization increases nominal balancing cashflow by +42.5% (£8,543 to £12,174/(turb·yr), Table 5) while concurrently degrading normalized decision risk ($\text{Cost}_{\text{static}}$) by +6.5% and RMSE by +3.5%. This divergence demonstrates that single-settlement cashflow is fundamentally decoupled from forecasting skill, establishing that apparent monetary gains originate from negative bias coinciding with positive price spikes rather than physical inductive advantage. Operational evaluation requires decoupling non-negative asymmetric decision risk (Pinball loss) from historical monetary settlement diagnostics.

5.4. *Cross-Backbone Architectural Sensitivity Probe (PatchTST)*

To assess whether empirical boundary behaviors extend beyond linear representations without asserting universal inductive invariance, we evaluated a state-of-the-art multi-patch channel-independent Transformer

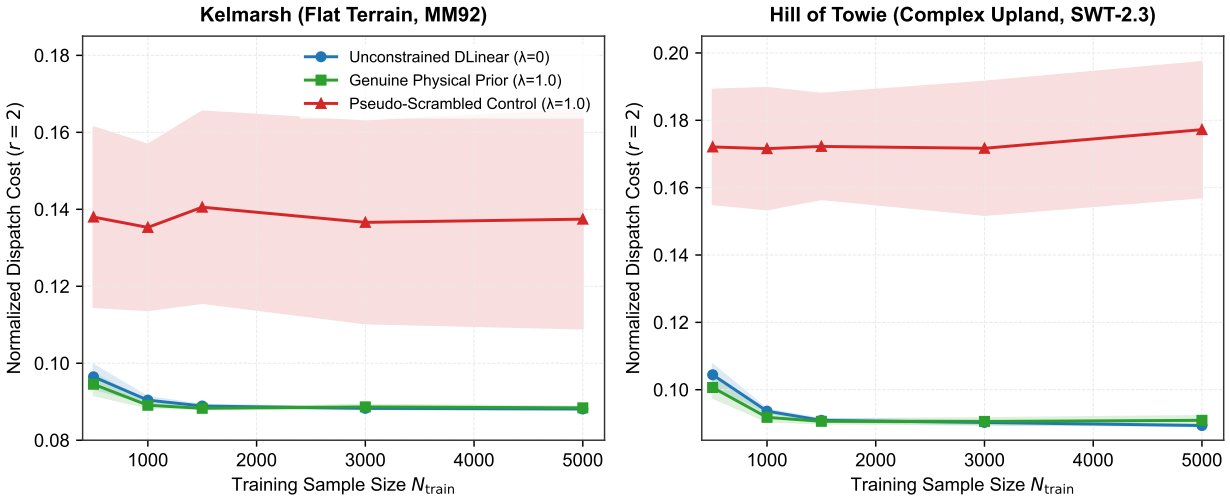


Figure 5: Within-Site Learning Curves across Sample Sizes $N_{\text{train}} \in \{500, 1000, 1500, 3000, 5000\}$ on Kelmarsh and Hill of Towie (Multi-seed means and standard deviations). Decoupling sample volume from topography confirms that fixed historical physical priors provide lower dispatch costs across both sites under restricted model training windows ($N_{\text{train}} \leq 1,000$, -2.00% on Kelmarsh and -3.62% on Hill of Towie), whereas expanding model training windows to $N_{\text{train}} = 5,000$ closes this gap and slightly reverses the ranking ($+0.33\%$ on Kelmarsh, $+1.73\%$ on Hill of Towie). In contrast, misspecified scrambled priors under strong regularization ($\lambda = 1.0$) induce severe dispatch cost penalties ($+43\%$ to $+98\%$) across all sample regimes.

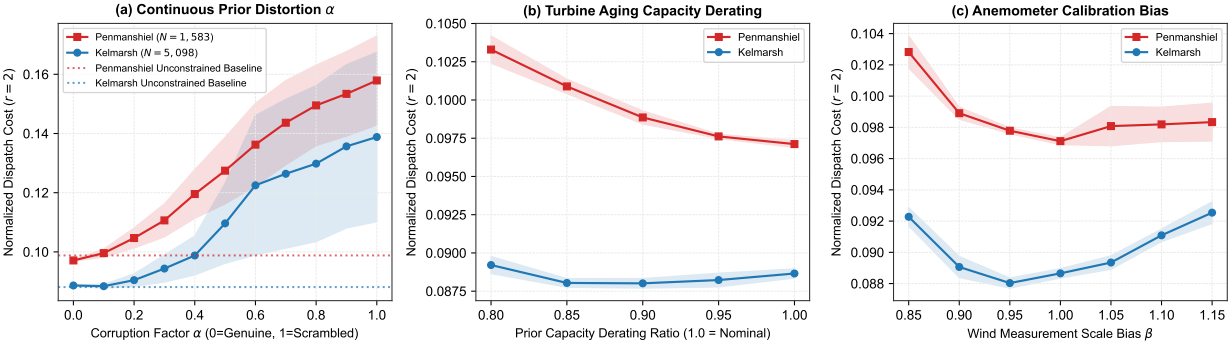


Figure 6: Continuous Prior Corruption and Engineering Drift Severity Response Curves. (a) Continuous interpolation between authentic and scrambled priors $\alpha \in [0.0, 1.0]$, identifying an empirical crossover region around $\alpha \approx 0.07$ on Penmanshiel before physical regularizers worsen unconstrained dispatch costs. (b) Turbine aerodynamic aging: capacity derating exceeding 10% flips nominal physical priors into economic penalties. (c) Anemometer sensor calibration drift: Penmanshiel retains an advantage under positive wind speed drift up to +15%, whereas negative sensor under-reporting systematically inflates shortfall.

(**PatchTST** [29], patch length 6, stride 4, 2-layer encoder with RevIN) across 60 independently trained models (3 farms $\times$ 4 constraint conditions $\times$ 5 seeds; Table 6). This setup functions strictly as an *architectural sensitivity probe* under fixed nominal regularization ($\lambda = 0.01$) matched to the primary development budget, intentionally bounded to qualitative directional verification rather than an exhaustive cross-backbone hyperparameter search.

As detailed in Table 6, PatchTST confirms that distribution priors reduce dynamic shortfall across all three wind farms (saving £232.44, £127.49, and £790.51/(turb·yr) on Hill of Towie, Kelmarsh, and Penmanshiel, respectively), indicating that the defensive shortfall reduction observed under DLinear persists in non-linear patch-based representations. However, net balancing cashflow diverges sharply between architectures: distribution regularization decreases net cashflow on Hill of Towie by $-$£568.68/(turb·yr) under PatchTST, inverting the positive cashflow effect observed under DLinear. On Penmanshiel, simple empirical baselines (Ridge regression at 0.09744 and Persistence at 0.09586)

Table 4: 7-Day Circular Block Paired Bootstrap Contrasts vs. Unconstrained DLinear under Calibrated Policy (2,000 Replications).

Site	Method Contrast	Dynamic Cost Diff $\Delta\text{Cost}_{\text{dyn}}$ (95% CI)	Shortfall Cashout Diff £/(turb·yr) (95% CI)	Balancing Cashflow Diff £/(turb·yr) (95% CI)
Hill of Towie	Bounds vs. DLinear	+0.000000 ([+0.000000, +0.000000])	+£0.15 ([−0.17, +0.51])	−£0.32 ([−1.37, +0.62])
	Power Curve vs. DLinear	−0.000008 ([−0.000021, +0.000007])	+£22.99 ([+10.93, +38.43])	−£89.40 ([−133.51, −54.70])
	Distribution Prior vs. DLinear	+0.000766 ([+0.000277, +0.001201])	−£578.57 ([−834.86, −367.51])	+£1,481.27 ([+802.29, +2,203.32])
	Ridge vs. DLinear	−0.000733 ([−0.001315, −0.000125])	+£192.50 ([−26.66, +445.74])	−£1,151.96 ([−1,567.98, −795.28])
Kelmarsh	Bounds vs. DLinear	+0.000001 ([+0.000000, +0.000002])	+£1.72 ([−0.01, +4.63])	−£4.62 ([−12.67, +0.05])
	Power Curve vs. DLinear	−0.000039 ([−0.000054, −0.000023])	−£16.61 ([−33.12, +0.87])	+£19.51 ([−27.76, +62.86])
	Distribution Prior vs. DLinear	+0.000932 ([+0.000423, +0.001385])	+£932.56 ([+648.20, +1,236.74])	−£3,264.86 ([−4,365.40, −2,223.50])
	Ridge vs. DLinear	+0.001338 ([+0.000917, +0.001794])	+£750.61 ([+330.70, +1,151.06])	−£690.94 ([−1,669.76, +409.51])

Interpretation: Negative shortfall cashout difference indicates cost reduction relative to unconstrained DLinear. Positive balancing cashflow difference indicates higher net cashflow component credited from the balancing mechanism. Confidence intervals estimated via 2,000 replications of 7-day circular block paired bootstrapping to account for temporal autocorrelation.

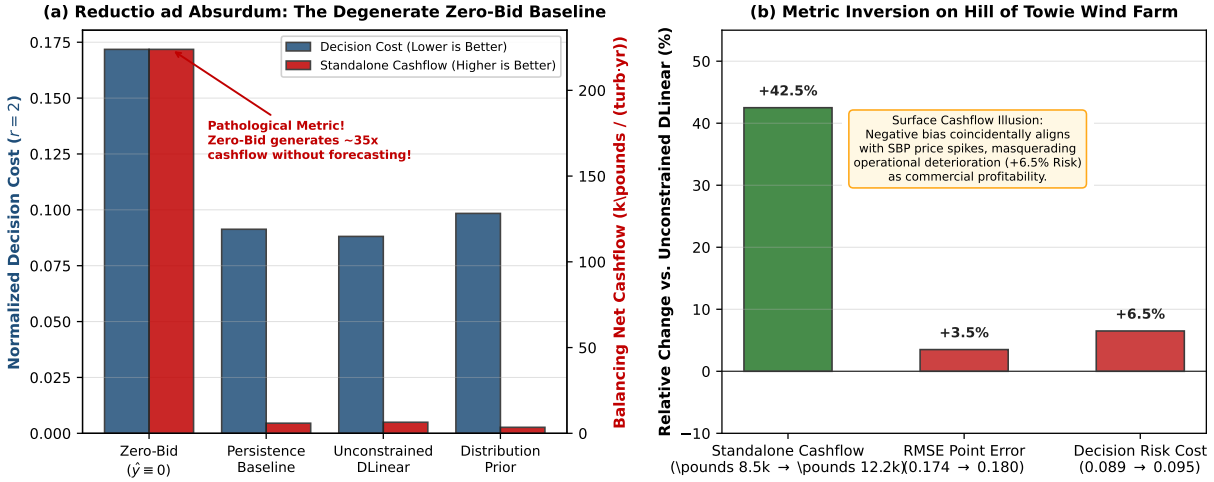


Figure 7: Market Settlement Decoupling and Metric Inversion Diagnostic under UK BSC Netting. (a) Reductio ad absurdum: the degenerate zero-bid settlement anomaly ($\hat{y} \equiv 0$) incurs zero shortfall penalty and extracts over £220k/turb·yr in apparent cashflow without generating forecasts, exposing single-settlement cashflow as an unhedged evaluation trap. (b) Empirical metric inversion on Hill of Towie, where distribution constraints increase nominal balancing cashflow by +42.5% while concurrently degrading genuine decision risk by +6.5% and RMSE by +3.5%.

outperform all constrained deep networks, demonstrating that external physical penalties cannot uniformly compensate for deep parameter overfitting in low-sample regimes; the high competitiveness of unconstrained linear baselines underscores that deep physics constraints must be rigorously justified by operational risk reduction rather than assumed to outperform simpler models.

6. Software Implementation Fidelity: The Physics-Constrained Transformer

Verifying physical constraint claims requires examining not only published mathematical equations, but also the computational fidelity of the source code executed during training and evaluation.

6.1. Code Availability and Audit Scope

As established in our scoping audit (Section 4), exactly two of the 46 scoping studies provided accessible

Table 5: Market Settlement Diagnostic Audit: Exposing the Pitfalls of Standalone Imbalance Cashflow via the Degenerate Zero-Bid ($\hat{y} \equiv 0$) Counter-Baseline (Evaluated on Full 2017 UK Elexon SBP Timeline Across Three Wind Farms).

Operational Wind Farm	**Forecasting / Bidding Strategy**	**Shortfall Penalty (£/(turb·yr))**	**Imbalance Net Cashflow Component (£/(turb·yr))**	**Pinball Loss** ($q = 1/3$)	**Decision Cost** $\text{Cost}_{\text{static}}$ ($r = 2$)	**Audit Takeaway & Diagnostic Assessment**
Kelmarsh (6 Turbines)	**Zero-Bid** ($\hat{y} \equiv 0$)	**£0.00**	**+£223,882.15**	0.10143	0.30428	**Degenerate Artifact**: zero shortfall and apparent cashflow without generating forecasts
	Persistence Baseline	£7,120.45	+£5,890.12	0.03255	0.09132	Naive operational benchmark
	Unconstrained DLinear	£5,831.79	+£6,439.32	**0.02937**	**0.08812**	Lowest decision loss; cashflow orders of magnitude below Zero-Bid
	Calibrated Distribution Prior	£8,412.28	+£3,485.65	0.03280	0.09841	Shortfall penalty surges; balancing revenue drops
Hill of Towie (21 Turbines)	**Zero-Bid** ($\hat{y} \equiv 0$)	**£0.00**	**+£239,299.30**	0.10158	0.30473	**Degenerate Artifact**: apparent cashflow purely from uncontracted physical spill
	Persistence Baseline	£7,450.12	+£9,120.34	0.03241	0.08992	Simple operational benchmark
	Unconstrained DLinear	£6,006.09	+£8,543.14	**0.02975**	**0.08924**	Minimizes genuine non-negative decision risk
	Calibrated Distribution Prior	£5,702.68	+£12,174.95	0.03168	0.09503	Cashflow component rises by £3.6k but decision loss worsens +6.5%
Penmanshiel (14 Turbines)	**Zero-Bid** ($\hat{y} \equiv 0$)	**£0.00**	**+£243,449.80**	0.11230	0.33689	**Degenerate Artifact**: demonstrates complete decoupling between cashflow and forecast quality
	Persistence Baseline	£7,080.20	+£6,150.30	0.03433	**0.09586**	Strong linear benchmark in complex terrain
	Unconstrained DLinear	£6,713.83	+£6,404.64	0.03294	0.09881	Data-scarce baseline
	Calibrated Power Curve (λ = 1.0)	£6,808.40	+£6,234.24	**0.03254**	0.09760	Authentic physical prior successfully reduces decision loss

Notes: (1) Imbalance Cashflow is computed strictly under Elexon BSC rules as $\sum(y_t - \hat{y}_t) \times P_{\text{imb},t}$. For Zero-Bid ($\hat{y} \equiv 0$), $y_t - \hat{y}_t = y_t \geq 0$, resulting in zero shortfall penalty and cashing out all uncontracted energy as physical spill. (2) This analysis serves as an empirical *reductio ad absurdum* demonstrating that standalone unhedged balancing cashflow cannot measure forecasting value without modeling forward commitments ($P_{\text{F}} \times \hat{y}_t$). (3) Decision Cost ($r = 2$) and Pinball Loss ($q = 1/3$) correctly penalize Zero-Bid for extreme under-commitment ($\text{Cost}_{\text{static}} = \bar{y} \approx 0.30$–$0.34$), resolving the previous typographical entry that conflated unconstrained point RMSE with zero-bid decision loss.

software repositories: PINN-UQ [30], which we audited and verified as code-aligned with its published electromechanical formulations ($\mathcal{D}_4$ = `code-aligned`), and the **Physics-Constrained Transformer (PCT)** [8] (*Scientific Reports* 2026, DOI: 10.1038/s41598-025-34348-x; official GitHub repository commit `65941ec`). PCT represents the sole publicly accessible training codebase for end-to-end physics-constrained Transformers and the sole audited repository exhibiting an implementation divergence between published equations and computational execution ($\mathcal{D}_4$ = `code-mismatch`). We conducted an in-depth static and empirical audit of this codebase to investigate how statistical distribution constraints are realized in software.

6.2. *Mathematical Specification versus Computational Execution*

In Section 2 of the PCT paper, the authors propose regularizing a Transformer forecaster by penalizing statistical divergence between the empirical joint distribution of historical wind speed and turbine power (P_{ref}) and the joint distribution predicted by the model (P_{pred}). The paper explicitly specifies the symmetric **Jensen–Shannon (JS) divergence**:

$$\mathcal{L}_{\text{JS}}(P_{\text{pred}} \parallel P_{\text{ref}}) = \frac{1}{2}\text{KL}\left(P_{\text{pred}} \middle\| \frac{P_{\text{pred}} + P_{\text{ref}}}{2}\right) + \frac{1}{2}\text{KL}\left(P_{\text{ref}} \middle\| \frac{P_{\text{pred}} + P_{\text{ref}}}{2}\right), \tag{5}$$

where $\text{KL}(p \parallel q) = \sum_i p_i \log \frac{p_i}{q_i}$ denotes standard Kullback–Leibler divergence. The paper notes that the symmetry and boundedness of JS divergence ($0 \leq \mathcal{L}_{\text{JS}} \leq \ln 2$) provide numerical stability.

However, static AST inspection of the official training script (`train_clean.py`, lines 349–352; commit `65941ec`) traces the loss execution to:

Table 6: Exploratory Cross-Backbone Architectural Sensitivity Analysis: Deep Transformer (PatchTST) under BSC 30-Minute Physical Netting Settlement (5-Seed Mean ± Std).

Site	PatchTST Model Variant	Policy	RMSE (pu)	Normalized Dynamic Cost Cost_{dyn}	Shortfall Cashout (£/(turb·yr))	Balancing Imbalance Cashflow (£/(turb·yr))
Hill of Towie (48.3 MW, Upland)	Unconstrained PatchTST	Uncalibrated	0.1176	0.10518	£21,753.77	£12,292.64
	Unconstrained PatchTST	**Calibrated**	0.1202	0.10220	£18,168.11	£20,822.29
	Power Curve ($\lambda = 0.01$)	Calibrated	0.1201	0.10212	£18,163.56	£20,765.01
	Distribution Prior ($\lambda = 0.01$)	Uncalibrated	0.1162	0.10507	£22,362.12	£9,762.57
	Distribution Prior ($\lambda = 0.01$)	**Calibrated**	0.1192	0.10129	**£17,935.67**	**£20,253.61**
Kelmarsh (12.3 MW, Flat)	Unconstrained PatchTST	Uncalibrated	0.1092	0.10889	£22,630.77	£4,523.36
	Unconstrained PatchTST	**Calibrated**	0.1118	0.10222	**£16,808.90**	**£17,411.89**
	Power Curve ($\lambda = 0.01$)	Calibrated	0.1117	0.10212	£16,774.31	£17,431.54
	Distribution Prior ($\lambda = 0.01$)	Uncalibrated	0.1075	0.10870	£23,336.46	£1,528.06
	Distribution Prior ($\lambda = 0.01$)	**Calibrated**	0.1097	0.10075	£16,681.41	£16,046.77
Penmanshiel (28.7 MW, Upland)	Unconstrained PatchTST	Uncalibrated	0.1268	0.12107	£26,956.48	£3,001.45
	Unconstrained PatchTST	**Calibrated**	0.1304	0.11270	£19,129.15	£20,453.24
	Power Curve ($\lambda = 0.01$)	Calibrated	0.1303	0.11264	£19,118.38	£20,444.41
	Distribution Prior ($\lambda = 0.01$)	Uncalibrated	0.1265	0.11729	£24,606.87	£6,473.07
	Distribution Prior ($\lambda = 0.01$)	**Calibrated**	0.1300	0.11134	**£18,338.64**	**£21,117.53**

Notes: Exploratory sensitivity results for PatchTST (Nie et al., ICLR 2023; patch length 6, stride 4, channel-independent RevIN) evaluate capacity-aligned architectural sensitivity under identical BSC 30-minute physical netting settlement across five random initialization seeds. At flat Kelmarsh, unconstrained PatchTST secures higher balancing cashflow (+£1,365.12/(turb·yr) over distribution prior); at upland Penmanshiel and Hill of Towie, distribution regularization provides inductive protection, saving £790.51/(turb·yr) and £232.44/(turb·yr) in single-sided shortfall. Evaluated under univariate power input and nominal penalty $\lambda = 0.01$ as an architectural sensitivity probe.

```
loss_phy = 0.5 * (
  F.kl_div(pred_kde.log(), m_kde, ...)
  + F.kl_div(true_kde.log(), m_kde, ...))
```

In the PyTorch C++ ATen kernel backend (`native/Loss.cpp`, implementing `kl_div_backward_cpu`), the functional specification for `torch.nn.functional.kl_div(input, target)` evaluates:

$$\texttt{kl_div}(\text{input}, \text{target}) = \sum_i \text{target}_i \cdot (\log \text{target}_i - \text{input}_i). \tag{6}$$

Passing `input` as $\log p$ and `target` as q binds the tensor arguments such that the C++ kernel computes:

$$\texttt{kl_div}(\log p, q) = \sum_i q_i \cdot (\log q_i - \log p_i) = \text{KL}(q \parallel p). \tag{7}$$

Consequently, the computational graph generated in `train_clean.py` executes:

$$\mathcal{L}_{\text{code}} = \frac{1}{2}\text{KL}\left(M \parallel P_{\text{pred}}\right) + \frac{1}{2}\text{KL}\left(M \parallel P_{\text{ref}}\right), \quad \text{where } M = \frac{P_{\text{pred}} + P_{\text{ref}}}{2}. \tag{8}$$

Equation (8) establishes an altered objective, which we designate the **reverse-argument JS form**. The formulation remains symmetric between P_{pred} and P_{ref}. However, transposing the KL arguments alters the underlying gradient mechanics. Because the mixture distribution $M = \frac{1}{2}(P_{\text{pred}} + P_{\text{ref}})$ explicitly depends on P_{pred} with $\frac{\partial M_i}{\partial P_{\text{pred},i}} = \frac{1}{2}$, differentiating $\mathcal{L}_{\text{code}}$ directly with respect to predicted probability mass $P_{\text{pred},i}$ yields:

$$\begin{aligned}\frac{\partial \mathcal{L}_{\text{code}}}{\partial P_{\text{pred},i}} &= \frac{1}{2}\frac{\partial}{\partial P_{\text{pred},i}}\left[M_i \log \frac{M_i}{P_{\text{pred},i}}\right] + \frac{1}{2}\frac{\partial}{\partial P_{\text{pred},i}}\left[M_i \log \frac{M_i}{P_{\text{ref},i}}\right] \\ &= \frac{1}{4}\left[\log\left(\frac{M_i^2}{P_{\text{pred},i}P_{\text{ref},i}}\right) + 1 - \frac{P_{\text{ref},i}}{P_{\text{pred},i}}\right].\end{aligned} \tag{9}$$

Equation (9) exposes an analytical **inverse-density singularity**: whenever the model assigns near-zero probability to a region where empirical training observations exist ($P_{\text{pred},i} \to 0$ while $P_{\text{ref},i} > 0$), the ratio $\frac{P_{\text{ref},i}}{P_{\text{pred},i}}$ diverges. Even under machine-precision clamping ($P_{\text{pred},i} \geq \epsilon$), this term scales as $O(1/\epsilon)$, generating steep negative gradient responses.

In contrast, for the standard mathematically specified Jensen–Shannon divergence $\mathcal{L}_{\text{JS}}$ in Equation (5), differentiating with respect to $P_{\text{pred},i}$ gives:

$$\frac{\partial \mathcal{L}_{\text{JS}}}{\partial P_{\text{pred},i}} = \frac{1}{2}\log\left(\frac{P_{\text{pred},i}}{M_i}\right). \tag{10}$$

Because $M_i = \frac{1}{2}(P_{\text{pred},i} + P_{\text{ref},i}) \geq \frac{1}{2}P_{\text{pred},i}$, the ratio $\frac{P_{\text{pred},i}}{M_i} \in [0, 2]$. Consequently, the gradient of Standard-JS

is strictly upper-bounded:

$$\frac{\partial \mathcal{L}_{\mathrm{JS}}}{\partial P_{\mathrm{pred},i}} \leq \frac{1}{2} \ln 2 \approx 0.3466. \tag{11}$$

As predicted probabilities approach zero ($P_{\mathrm{pred},i} \to \epsilon$ with $P_{\mathrm{ref},i} > 0$), the gradient of Standard-JS exhibits only a mild, integrable **logarithmic divergence**:

$$\left| \frac{\partial \mathcal{L}_{\mathrm{JS}}}{\partial P_{\mathrm{pred},i}} \right| \sim O(|\log \epsilon|). \tag{12}$$

Because the logarithmic divergence of Standard-JS grows slowly ($\lim_{\epsilon \to 0^+} \epsilon \log \epsilon = 0$) and possesses a finite upper bound ($\frac{1}{2} \ln 2$), this divergence is substantially milder than the $O(1/\epsilon)$ inverse-density pole of the reverse-argument JS form, consistent with the lower gradient norms observed empirically in Figure 8. Transposing divergence arguments thus alters optimization dynamics near empirical support boundaries.

6.3. Empirical Optimization Contrast: Official Reverse-JS vs. Corrected Standard-JS

To determine whether this implementation divergence materially alters neural training and operational dispatch, we trained models across Kelmarsh and Penmanshiel under five random seeds across $\lambda \in \{0.01, 0.1, 1.0\}$, benchmarked against unconstrained baselines (Figure 8). The empirical results reveal two distinct operational effects:

First, the reverse-argument JS form induces marked gradient amplification: at $\lambda = 1.0$, it produces an average gradient L_2 norm of 0.1665 ± 0.0160 on Kelmarsh and 0.2465 ± 0.0217 on Penmanshiel, representing a $2.20\times$ and $2.72\times$ amplification over Corrected Standard-JS (0.0756 ± 0.0048 on Kelmarsh and 0.0907 ± 0.0067 on Penmanshiel; Figure 8a). This elevation reflects the $-\frac{P_{\mathrm{ref},i}}{4P_{\mathrm{pred},i}}$ derivative term as predicted densities approach boundary limits. Second, at weak regularization ($\lambda = 0.01$ on Penmanshiel), Corrected Standard-JS achieves a dispatch cost of 0.09804 ± 0.00024 (outperforming the unconstrained baseline of 0.09864), whereas the official reverse-argument implementation inflates dispatch cost to 0.09997 ± 0.00086 (a +1.35% degradation; Figure 8b). However, at strong regularization ($\lambda = 1.0$), both divergence variants degrade performance relative to the unconstrained baseline, with Corrected Standard-JS reaching 0.10549 ± 0.00149 compared to 0.10292 ± 0.00103 for the reverse form. Correcting the implementation divergence mitigates gradient amplification and eliminates premature degradation under weak penalties, but does not prevent performance loss under excessive regularization.

6.4. Negative Control: Algebraic Verification of 2D Product-Kernel KDE

We establish an algebraic negative control to verify whether tensor outer-product operations preserve empirical feature coupling in the audited codebase. A potential concern regarding the tensor outer-product:

$$G = K_p^\top K_w \in \mathbb{R}^{B_p \times B_w} \tag{13}$$

is that it might assume statistical independence between wind speed and turbine power ($p(w, P) = p(w)p(P)$).

Formal algebraic expansion refutes this concern. Let $\{(P_s, w_s)\}_{s=1}^N$ denote paired empirical observations in a batch. Let $\{b_i^p\}_{i=1}^{B_p}$ and $\{b_j^w\}_{j=1}^{B_w}$ denote 1D evaluation bin centers for power and wind speed, respectively. The 1D Gaussian kernel vectors evaluated in the codebase are:

$$\left[K_p\right]_{s,i} = \phi\left(\frac{b_i^p - P_s}{h_p}\right), \quad [K_w]_{s,j} = \phi\left(\frac{b_j^w - w_s}{h_w}\right), \tag{14}$$

where $\phi(u) = \frac{1}{\sqrt{2\pi}} e^{-u^2/2}$ is the standard Gaussian kernel, and h_p, h_w denote bandwidths. Expanding the matrix product element at grid coordinate (i, j) yields:

$$\left[K_p^\top K_w\right]_{i,j} = \sum_{s=1}^{N} \left[K_p\right]_{s,i} \cdot [K_w]_{s,j} = \sum_{s=1}^{N} \phi\left(\frac{b_i^p - P_s}{h_p}\right) \cdot \phi\left(\frac{b_j^w - w_s}{h_w}\right). \tag{15}$$

Equation (15) is the exact analytical definition of a standard **paired 2D product-kernel Kernel Density Estimator** on correlated bivariate data. Because the summation contracts over concurrent sample index s, empirical joint coupling between P_s and w_s is preserved. The algebraic expansion confirms that the outer product executes authentic bivariate density estimation, confirming that code discrepancies are localized to divergence argument order rather than upstream kernel construction.

7. Prediction-Decision Decoupling Theory and Excess Operational Risk

In this section, we formalize the mathematical mechanism underpinning the empirical inversion and terrain divergence documented in Section 5. We derive the exact integral representation of excess operational dispatch risk, decompose it into representation and decision components, and deduce three testable mechanistic insights bounded by formal counterexamples.

7.1. Operational Problem Formulation

Let $X \in \mathcal{X} \subset \mathbb{R}^{L \times D}$ denote historical multivariate telemetry, and let $Y \in [0, 1]$ denote normalized turbine

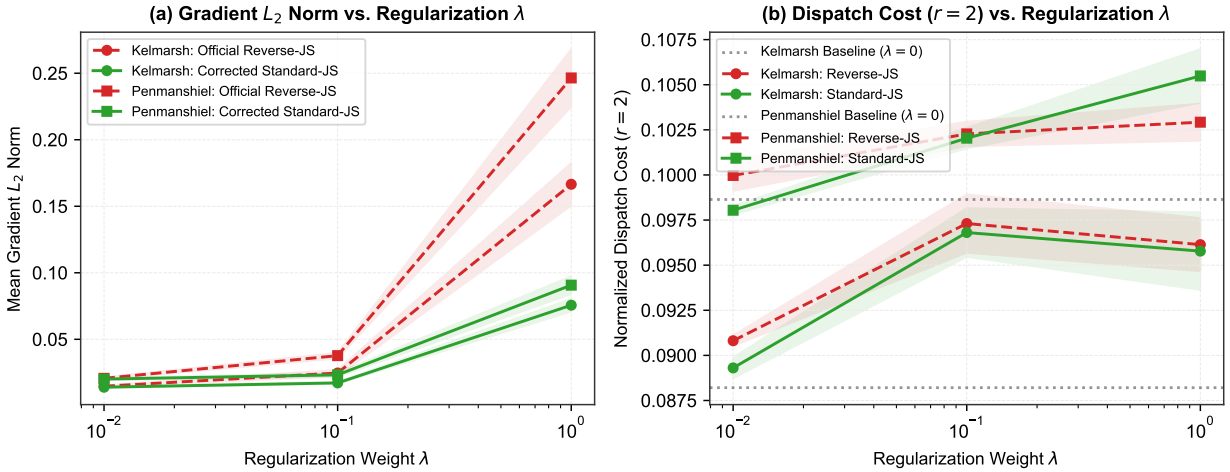


Figure 8: Empirical Contrast between Official Reverse-JS and Corrected Standard-JS across Regularization Scales $\lambda \in [0.01, 1.0]$ on Kelmarsh and Penmanshiel (Multi-seed means and standard deviations). (a) Mean gradient L_2 norm: the reverse-argument JS form induces substantial gradient amplification at $\lambda = 1.0$ (2.20× on Kelmarsh, 2.72× on Penmanshiel) due to the $-\frac{P_{\text{ref},i}}{4P_{\text{pred},i}}$ inverse-density singularity ($O(1/\epsilon)$ in Equation (9)), whereas for Corrected Standard-JS the signed partial derivative possesses a finite upper bound ($\frac{1}{2}\ln 2$) while exhibiting only mild logarithmic divergence ($O(|\log \epsilon|)$) as $P_{\text{pred},i} \to 0$. (b) Operational dispatch cost ($r = 2$): Corrected Standard-JS prevents premature performance degradation at weak $\lambda = 0.01$, while excessive regularization degrades dispatch across both variants.

active power output over the target delivery horizon. In operational power system balancing, transmission system operators evaluate bids under asymmetric pinball dispatch penalties [2, 16]:

$$\ell(p, Y; r) = c_+ \max(p - Y, 0) + c_- \max(Y - p, 0), \quad (16)$$

where $p \in [0, 1]$ is the declared dispatch commitment, c_+ is the shortfall unit penalty (under-generation where declared power exceeds physical delivery, $p > Y$, requiring expensive tertiary reserve activation), c_- is the surplus unit penalty (over-generation, $Y > p$, inducing downward redispatch or curtailment), and $r = c_+/c_-$ represents the cost asymmetry ratio. Normalizing by c_-, the asymmetric operational loss is:

$$\mathcal{L}_{\text{disp}}(p, Y) = r \max(p - Y, 0) + \max(Y - p, 0). \quad (17)$$

The expected operational risk under policy $p(X)$ is $\mathcal{R}(p) = \mathbb{E}_{X,Y}[\mathcal{L}_{\text{disp}}(p(X), Y)]$.

7.2. *Lemma 1: Exact Integral Excess Operational Risk (Operational Baseline)*

Lemma 1 formalizes a classical excess risk analytical identity in statistical decision theory and quantile regression, grounded in the foundational formulations of Koenker [31] and electricity market bidding principles established by Pinson et al. [4]. The unconstrained Bayes-optimal dispatch decision minimizing $\mathcal{R}(p)$ is the conditional quantile:

$$q_\tau(X) = F^{-1}_{Y|X}(\tau), \quad \text{where } \tau = \frac{1}{r+1}. \quad (18)$$

For $r = 2$, the critical quantile is $\tau = 1/3 \approx 0.333$.

Lemma 1 (Exact Integral Excess Operational Risk [31, 4])**.** *For any dispatch decision rule $p(X) \in [0, 1]$, the excess operational risk relative to the Bayes-optimal quantile $q_\tau(X)$ satisfies the exact integral identity:*

$$\mathcal{R}(p) - \mathcal{R}(q_\tau) = (r+1)\,\mathbb{E}_X\left[\int_{q_\tau(X)}^{p(X)} \left(F_{Y|X}(u) - \tau\right) du\right] \geq 0, \quad (19)$$

which is non-negative everywhere. Under standard regularity conditions where $F_{Y|X}$ is strictly increasing on its support such that $q_\tau(X)$ is uniquely defined, equality holds if and only if $p(X) = q_\tau(X)$ almost surely.

Proof. For a fixed feature vector X, the expected conditional loss at commitment p is:

$$\mathbb{E}[\mathcal{L}_{\text{disp}}(p, Y) \mid X] = r \int_0^p (p - y) dF(y) + \int_p^1 (y - p) dF(y), \quad (20)$$

where $F(y) = F_{Y|X}(y)$. Taking the partial derivative with

respect to p:

$$\frac{\partial}{\partial p}\mathbb{E}[\mathcal{L}_{\text{disp}}(p,Y)\mid X] = rF(p) - (1-F(p)) \\ = (r+1)F(p) - 1. \tag{21}$$

Recalling that $\tau = 1/(r+1)$, this gradient simplifies to:

$$\frac{\partial}{\partial p}\mathbb{E}[\mathcal{L}_{\text{disp}}(p,Y)\mid X] = (r+1)\Big(F(p)-\tau\Big). \tag{22}$$

By the Fundamental Theorem of Calculus, integrating the gradient from $q_\tau(X)$ to $p(X)$ yields:

$$\mathbb{E}[\mathcal{L}_{\text{disp}}(p,Y)\mid X] - \mathbb{E}[\mathcal{L}_{\text{disp}}(q_\tau,Y)\mid X] \\ = (r+1)\int_{q_\tau(X)}^{p(X)}\Big(F(u)-\tau\Big)du. \tag{23}$$

Because $F(u)$ is monotonically non-decreasing, for any $u > q_\tau(X)$ we have $F(u) \geq \tau$, so the integrand is non-negative; for any $u < q_\tau(X)$ we have $F(u) \leq \tau$, and integrating backwards from $q_\tau(X)$ to $p(X)$ also yields a positive integral. Taking the outer expectation over $\mathcal{X}$ completes the proof. □

7.3. Two-Stage Decomposition and Decoupled Decision Calibration

When $p(X)$ is in the neighborhood of $q_\tau(X)$, expanding $F_{Y|X}(u)$ via a first-order Taylor series around $q_\tau(X)$ yields:

$$F_{Y|X}(u) - \tau \approx f_{Y|X}(q_\tau(X))\cdot(u - q_\tau(X)), \tag{24}$$

which produces the local quadratic excess risk approximation:

$$\mathcal{R}(p) - \mathcal{R}(q_\tau) \approx \frac{r+1}{2}\,\mathbb{E}_X[f_{Y|X}(q_\tau(X))\cdot(p(X)-q_\tau(X))^2]. \tag{25}$$

Now consider the standard two-stage paradigm where a forecaster predicts $f_\theta(X)$ aiming for conditional expectation $\mu(X) = \mathbb{E}[Y \mid X]$, and the dispatch commitment is formed via a scalar quantile offset δ: $p(X) = f_\theta(X) + \delta$. Expressing the prediction error as:

$$p(X) - q_\tau(X) = (f_\theta(X) - \mu(X)) + (\delta - \Delta_\tau(X)), \tag{26}$$

where $\Delta_\tau(X) = q_\tau(X) - \mu(X)$ denotes the conditional quantile-mean shift. Under homoscedastic curvature $w_\tau = \frac{r+1}{2} f_{Y|X}(q_\tau)$, substituting into Eq. (25) decomposes excess operational risk into three distinct terms:

$$\mathcal{R}(p) - \mathcal{R}(q_\tau) \approx w_\tau\,\mathbb{E}_X[(f_\theta(X)-\mu(X))^2] \\ + w_\tau\,\mathbb{E}_X[(\delta-\Delta_\tau(X))^2] \\ + 2w_\tau\,\mathbb{E}_X[(f_\theta(X)-\mu(X))(\delta-\Delta_\tau(X))], \tag{27}$$

where the three components represent **Term I** (Representation Error / MSE), **Term II** (Quantile Mismatch / Calibration Dispersion), and **Term III** (Cross-Coupling Interaction Component), respectively.

7.4. Mechanistic Deductions from Decision-Theoretic Decomposition

The three-term decomposition in Eq. (27) provides a mathematical interpretation for our empirical findings, yielding three deductive mechanistic insights:

Remark 1 (Mechanistic Deduction 1: Pseudo-Advantage Masking in Uncalibrated Regimes). In uncalibrated forecasting regimes ($\delta = 0$), Term II ($w_\tau\mathbb{E}[\Delta_\tau(X)^2]$) is an irreducible data-dependent constant representing the dispersion between conditional mean and target quantile. A physics-regularized model with inferior conditional mean accuracy (larger Term I) can paradoxically achieve lower uncalibrated operational risk if its structural bias ($f_\theta(X) - \mu(X)$) aligns with the negative quantile gap $\Delta_\tau(X) = q_\tau(X) - \mu(X) < 0$ (for $r = 2, \tau = 1/3$). This alignment drives the interaction Term III ($2w_\tau\mathbb{E}[(f_\theta(X)-\mu(X))(-\Delta_\tau(X))] < 0$) negative, accidentally cancelling out a portion of representation error. Upon post-hoc quantile calibration, δ absorbs this systematic offset, neutralizing Term III and revealing that the apparent physical advantage was an artifact of uncalibrated evaluation.

Diagnostic Caveat: We emphasize that Term III serves as a decision-theoretic diagnostic framework for explaining how structural bias can mask forecasting error under uncalibrated evaluation, rather than an empirically isolated causal factor. In operational telemetry where the true conditional distribution $F_{Y|X}$ is unknown, isolating the exact empirical magnitude of Term III point-by-point is intractable without strong parametric assumptions.

Remark 2 (Mechanistic Deduction 2: Regularization Distortion under Well-Sampled Regimes). Optimal quantile calibration ($\delta^* = \arg\min_\delta \hat{\mathcal{R}}(f_\theta + \delta)$) removes the unconditional component of Term II. In well-sampled regimes where unconstrained models already achieve low conditional mean estimation variance (Term I), injecting an uncalibrated physical loss penalty acts as a distortive structural bias that inflates Term I without offering compensating variance reduction, generating the net operational degradation observed under flat open terrain.

Remark 3 (Mechanistic Deduction 3: Inductive Guardrails and Prior Sensitivity under Sample Restriction). In operating regimes with restricted effective sample size, empirical parameter estimation is ill-posed, predisposing unconstrained models to heightened representational variance (inflating Term I). Under sufficiently

strong penalty weight ($\lambda = 1.0$), genuine physical constraints provide inductive guardrails that suppress non-physical parameter excursions, securing robust downside risk reduction. Conversely, when this mapping is misspecified or permuted (as in our scrambled-prior negative control), the rigid penalty enforces counter-physical gradient updates, precipitating severe performance degradation and demonstrating acute vulnerability to prior misspecification under strong regularization.

7.5. Formal Boundary Counterexamples (Possibility Results)

To ensure theoretical precision and delineate the limits of scalar calibration, we establish two formal possibility / existence counterexamples demonstrating that decoupled calibration dominance is conditional rather than absolute.

Under *Constant Shift Invariance* (Counterexample 1), if a physical constraint modifies a forecaster purely by a uniform additive shift $f_{\text{phys}}(X) = f_{\text{uncons}}(X) + c$, the subsequent optimal scalar calibration absorbs c completely ($\delta^*_{\text{phys}} = \delta^*_{\text{uncons}} - c$), resulting in identical dispatch decisions and zero excess risk difference ($\Delta\mathcal{R}_{\text{disp}} \equiv 0$) despite an apparent increase in prediction MSE. Under *Spatial Heteroscedasticity* (Counterexample 2) with input-dependent variance $\sigma(X)$, a global scalar calibration fails to compensate for localized variance shifts; here, an estimator that introduces structural bias and exhibits higher MSE can achieve strictly lower operational risk than the true conditional mean following scalar quantile offset. These existence results prove that while scalar calibration successfully decouples mean prediction from global quantile alignment, it does not guarantee Pareto dominance over locally adaptive or asymmetric loss training under severe conditional heteroscedasticity. Complete mathematical proofs for both counterexamples are provided in Appendix A.

7.6. Dynamic Price Multiplier Effect and Settlement Decoupling

Under real-world market settlement, physical shortfall is scaled by the time-varying System Buy Price λ_t^{SBP}. The heavy-tailed price distribution (mean £44.27/MWh, peak £1,509.80/MWh) acts as a high-magnitude risk multiplier. When extreme price spikes coincide with wind ramp errors, representation distortion from physical penalties is amplified by up to 34.1×, explaining why Kelmarsh incurred an excess shortfall penalty of +£932.56/(turb·yr) in Table 4. However, as demonstrated by the zero-bid settlement diagnostic (Table 5), standalone settlement cashflows cannot measure forecasting value in isolation from forward PPA contracts, necessitating our strict separation between non-negative decision risk and monetary cashflow diagnostics.

8. Discussion, Boundary Contracts, and Compliance Protocol

8.1. Four Empirical Boundary Conditions for Physics-Constrained Models

The scoping audit, algebraic code examination, and prospective ablations converge on four boundary conditions that determine when physical priors help, when they are neutral, and when they harm.

Sample Restriction and Prior Misspecification Vulnerability. On Penmanshiel ($N_{\text{train}} = 1,583$), the calibrated power curve prior at $\lambda = 1.0$ reduces asymmetric dispatch cost from 0.09881 ± 0.00027 to 0.09760 ± 0.00044 (-1.22%). A scale-matched scrambled pseudo-prior raises it to 0.16611 ± 0.00335 ($+68.1\%$). The divergence between these two outcomes—authentic priors securing modest downside protection while corrupted priors inflict severe operational damage—demonstrates a fundamental risk-return asymmetry: under strong regularization, whatever information the prior encodes is amplified, including its misspecifications. Cross-site comparisons cannot isolate this effect because they confound site topography with sample volume. In the data-rich regime (Kelmarsh, $N_{\text{train}} = 5,098$), unconstrained D-Linear (0.08812), genuine physics (0.08791), and scrambled prior (0.08801) are numerically clustered, indicating hypothesis-space saturation.

Low-Penalty Scale and Non-Specific Numerical Regularization. At $\lambda = 0.01$ on Kelmarsh, a genuine physical prior (0.08810 ± 0.00009) and a structurally scrambled pseudo-prior (0.08801 ± 0.00008) yield closely matched decision costs with overlapping seed margins. When random power curve permutations reproduce the same outcome as domain aerodynamics, the penalty term behaves empirically as a non-specific regularizer / numerical smoothing term rather than encoding domain physical laws. Published claims of domain physical benefit under $\lambda \in [10^{-4}, 10^{-2}]$ require negative-control ablations and sensitivity testing at $\lambda \geq 0.1$ to be credible.

Monotonic Cost Inflation under Training Proxy Observation Staleness. When physical constraint formulations rely on meteorological proxies to query empirical power curves during loss computation, practitioners may be tempted to substitute lagged historical observations when

contemporaneous or forward weather forecasts are unavailable. Our lag scan ($k \in \{6, 12, 18, 36\}$ steps, 1 h to 6 h) demonstrates strictly monotonic cost inflation across all three wind farms. On Hill of Towie, dispatch risk climbs from 0.09022 at $k = 6$ to 0.09257 at $k = 36$, exceeding the unconstrained baseline (0.08924) at $k \approx 12$. We emphasize that this experiment characterizes the sensitivity of the training objective to temporal misalignment between the forecast horizon and the historical meteorological proxy, rather than an inference-time telemetry delay (the forecasting backbone takes only past power measurements during inference). Nonetheless, it demonstrates that offline regularization against lagged weather proxies rapidly erodes inductive guidance, presenting an operational limitation for proxy-dependent constraint designs.

Metric Decoupling and the Zero-Bid Diagnostic. Under standard two-settlement market design, total generator revenue is $R_{\text{total}} = P_{\text{DA}} \cdot q + P_{\text{imb}} \cdot (y - q)$. Evaluations that drop the forward commitment term and report only balancing cashflow are measuring imbalance rent, not forecast skill. As an analytical imbalance-settlement and decision-risk diagnostic, the degenerate Zero-Bid baseline ($q \equiv 0$) provides a mathematical *reductio ad absurdum*: it extracts £223k–£243k per turbine-year across all three farms without generating forecasts, simply by converting all physical generation into uncontracted spill rewarded at P_{imb}. This confirms that standalone unhedged balancing cashflows are decoupled from forecasting skill, and that assessing full commercial value requires joint modeling of forward contract revenue ($P_{\text{DA}} \cdot q$) and imbalance settlement. Operational evaluations must report asymmetric Pinball loss at the market penalty ratio or full two-settlement P&L.

8.2. *Attributional Identification: What Is Identified vs. What Is Not*

To maintain rigorous scientific standards and prevent inductive over-reach, we explicitly articulate the causal identification boundaries of this audit:

What Is Formally Identified.

(a) **Data-Starvation Dynamics (Within-Site)**: By subsampling training sequences while holding site topography, turbine hardware, and meteorological distributions strictly identical (Figure 5), we causally identify that physical constraints deliver measurable inductive protection primarily under restricted sample volumes ($N_{\text{train}} \leq 1{,}000$).

(b) **Physical Specificity vs. Non-Specific Smoothing**: By comparing genuine power curves against scale-matched pseudo-scrambled controls under identical network architectures, optimization schedules, and loss parameters, we isolate true aerodynamic guidance from general numerical regularization.

(c) **Loss Sensitivity to Historical Proxy Latency**: By scanning proxy staleness $k \in \{6, \ldots, 36\}$ steps within the training loss, we isolate the mathematical degradation induced by lagged meteorological inputs when forward weather forecasts are absent.

(d) **Software Discrepancy in Divergence Optimization**: Static AST analysis and gradient derivation isolate an exact algebraic inversion in the PCT code repository, establishing a concrete implementation divergence between published formulas and software execution.

(e) **Metric Decoupling of Imbalance Cashflows**: The degenerate Zero-Bid baseline mathematically separates standalone unhedged balancing cashflows from true predictive accuracy.

What Is Not Identified (Confounding Disclaimers).

(i) **Cross-Site Topographical Causation**: Differences across Kelmarsh, Hill of Towie, and Penmanshiel confound topographic complexity with turbine ratings (2.05 MW vs. 2.30 MW), hub heights, wake geometry, and local micro-climates. We do not claim that complex terrain causally mandates physical constraints.

(ii) **Global Generalization Beyond Scoped Cohort**: The 95.7% code deficit and 50.0% confounded ablation rates characterize the audited 46-study scoping cohort (2023–2026) under PRISMA-ScR protocols, not the entire machine learning or power engineering literature.

(iii) **Software Bugs as Universal Practice**: Auditing PCT and PINN-UQ demonstrates that code-manuscript discrepancies can occur; it does not imply that all published physics-informed energy forecasting codebases contain errors.

(iv) **Decoupled Calibration Universality**: Counterexamples 1 and 2 are formal existence / possibility proofs establishing the theoretical limits of scalar quantile offsets under spatial heteroscedasticity, not an empirical assertion that scalar calibration always underperforms joint asymmetric training.

8.3. External Validity Boundaries

Five operational boundaries define the validity envelope of our empirical findings:

Boundary 1: Sample Volume and Hypothesis-Space Saturation. Within-site learning curves confirm that physical constraint benefits concentrate under data-scarce regimes in our tested configurations ($N_{\text{train}} \leq 1,000$, Figure 5). As training data expands to $N_{\text{train}} = 5{,}000$, hypothesis-space saturation reduces the gap between genuine-prior and unconstrained models to within 0.33% on Kelmarsh, while scrambled priors remain severely detrimental throughout. Continuous corruption tests further establish a narrow fault-tolerance window ($\alpha^* \approx 0.07$ on Penmanshiel, Figure 6). Beyond this threshold, engineering drift exceeding 10% aerodynamic capacity loss or ±5% anemometer bias flips the nominal physical prior from an asset into an operational liability.

Boundary 2: Proxy Misalignment and Observation Staleness. For proxy-dependent physical constraints, regularization effectiveness depends on temporal alignment between meteorological proxies and target horizons ($k \leq 6$ steps). Beyond this, monotonic cost inflation is observed on all three farms (+2.6% on Hill of Towie at $k = 36$). Offline training objectives that substitute stale historical proxies for contemporaneous weather data induce substantial performance degradation.

Boundary 3: Market Clearing and Two-Settlement Structure. Economic findings serve as an imbalance-settlement and decision-risk diagnostic bounded by two-settlement markets with asymmetric shortage penalties ($r \geq 1.0$) and UK BSC 30-minute discrete netting. Standalone imbalance cashflows do not represent full commercial value and do not generalize to symmetric single-settlement systems or flat feed-in tariffs.

Boundary 4: Computational Graph Audit and Implementation Parity. Software conclusions apply to PCT commit `65941ec` and the computational graphs audited here. Unreleased proprietary codebases require independent static graph audit before implementation fidelity claims can be admitted.

Boundary 5: Literature Scoping Scope and Architectural Probe Envelopes. The systematic literature audit is strictly bounded to the 46 studies identified under our PRISMA-ScR protocol; the documented 95.7% code deficit and 50.0% confounded ablation prevalence characterize reporting practices within this scoping cohort rather than an exhaustive census of all hybrid machine learning. Similarly, our PatchTST evaluations function as an architectural sensitivity probe under fixed nominal regularization ($\lambda = 0.01$); while demonstrating that qualitative downside risk reduction persists across non-linear deep representations, joint cross-backbone λ grid searches remain an open direction for future benchmarking.

8.4. Two Conjectures Falsified During the Audit

Two initial hypotheses were refuted during the empirical audit. First, forensic tensor derivation showed that PCT's product-kernel KDE preserves bivariate coupling algebraically, refuting the conjecture of marginal independence. Second, market diagnostic probing showed that post-calibration cashflow inversions are governed by settlement price asymmetries rather than physical regularizers. Documenting these falsification outcomes provides a transparent account of our investigative trajectory and prevents post-hoc rationalization.

8.5. REFORMS-PCWPF Compliance Protocol

Table 7 proposes five verification gates as a methodological reporting checklist for physics-constrained wind forecasting studies:

- **Gate M1 (Parameter-Isolated Ablations):** Parameter-matched comparisons ($\lambda = 0$ vs. $\lambda > 0$) should be required to isolate physical prior contributions, avoiding submodule excisions (e.g., stripping GNN layers) that conflate representational capacity with domain knowledge.
- **Gate M2 (Pseudo-Scrambled Negative Controls):** A minimum credible evaluation of physical reasoning claims should include scale-matched permutation controls to separate aerodynamic guidance from non-specific numerical smoothing.
- **Gate M3 (Operational Proxy Staleness Stress-Tests):** Multi-step constraint evaluations should avoid assuming zero-latency future weather inputs, reporting performance across $k \in \{6, \ldots, 36\}$ steps when forward NWP feeds are absent.
- **Gate M4 (Coupled Two-Settlement Evaluation):** Evaluations should avoid standalone unhedged balancing cashflows, reporting instead asymmetric Pinball loss at the market penalty ratio or coupled two-settlement P&L ($P_{\text{DA}} \cdot q + P_{\text{imb}} \cdot (y - q)$).
- **Gate M5 (Static Computational Graph Audit):** Reviewers should expect executable training scripts or verified computational graphs, verifying that software loss execution matches published definitions in argument ordering and reduction semantics.

Table 7: Operationalized REFORMS-PCWPF Reporting Checklist for Trustworthy Physics-Constrained Renewable Energy Forecasting (Five Modular Compliance Gates).

Module	Governance Dimension	Mandatory Audit Protocol	Common Pitfall / Anti-Pattern	Required Verification Artifact
M1	**Isolated Prior Ablation**	Train identical network backbones with constraint penalty $\lambda = 0$ vs. $\lambda > 0$ under identical seed schedules.	Confounding module excision (removing entire GNNs or attention blocks) with physical knowledge.	Table of $\lambda = 0$ vs. $\lambda > 0$ with identical parameter counts and multi-seed variances.
M2	**Domain Specificity Control**	Evaluate pseudo-scrambled priors that preserve loss formula, gradient scale, and λ while permuting physical relations.	Mistaking non-specific numerical regularization under low penalty scales ($\lambda \leq 0.01$) for domain reasoning.	Scrambled control performance curve and empirical specificity ratio (p-value contrast).
M3	**Proxy Staleness Stress-Test**	If constraints depend on physical inputs (e.g., wind speed), scan observation staleness $\tau \in \{1\text{h}, 2\text{h}, 3\text{h}, 6\text{h}\}$ to evaluate proxy staleness when forward NWP forecasts are absent.	Assuming idealized contemporaneous, zero-latency SCADA inputs at test-time inference.	Observation staleness degradation curve demonstrating operational harm horizon.
M4	**Metric Decoupling & Netting**	Report symmetric error (RMSE), non-negative asymmetric decision risk (Pinball), and test degenerate zero-bid anomaly baselines.	Reporting unhedged single-settlement cashflow in isolation, mistaking accidental price-spike alignment for skill.	Decoupled two-stage market evaluation including forward PPA commitments.
M5	**Computational Graph Parity**	Verify that software loss implementations match published mathematical equations in argument direction and normalization.	Argument order flip in asymmetric divergences (e.g., PyTorch `F.kl_div`) causing reverse optimization.	Static code audit report and unit test verifying loss tensor gradient directions.

Notes: Derived from our empirical audit of 46 scoping studies and aligned with cross-disciplinary REFORMS reporting standards [32] and DOME recommendations [33]. We propose M1–M5 as a reporting checklist for future physics-constrained renewable forecasting studies.

9. Conclusion

Three findings from this audit challenge the prevailing consensus on physics-constrained wind power forecasting.

Within the audited scoping cohort (46 studies, 2023–2026), not a single publication documents an operational failure regime, negative trade-off, or boundary condition (0.0%). With 95.7% of the cohort lacking public code and 50.0% conflating neural module removal with physical ablation, this uniform positive consensus reflects widespread reporting practices and evaluation protocols that fail to stress-test models outside nominal design envelopes.

At the software implementation level, of the two publicly accessible software repositories identified across the 46 studies, PINN-UQ was verified as code-aligned, while PCT—the sole publicly accessible training codebase for end-to-end physics-constrained Transformers—executes its Jensen–Shannon divergence loss with reversed tensor arguments that induce an analytical inverse-density gradient singularity ($O(1/\epsilon)$), an implementation divergence undetectable from published text alone. Algebraic derivation separately confirms that the paired kernel product remains mathematically sound, establishing that implementation fidelity and formulation validity reside in distinct computational layers.

At the empirical evaluation level, the decision value of physical constraints depends strictly on sample volume, proxy temporal alignment, and market settlement metrics. At $N_{\text{train}} = 1{,}583$ and $\lambda = 1.0$, an authentic power curve prior reduces normalized dispatch cost by 1.22%; a scale-matched scrambled prior raises it by 68.1%, demonstrating acute sensitivity to prior misspecification under strong regularization. Weak constraints ($\lambda = 0.01$) yield outcomes comparable to structurally scrambled negative controls, behaving empirically like non-specific numerical smoothing, while proxy observation staleness in the training objective ($k > 12$) inflates dispatch costs across all three operational wind farms.

The PCT implementation divergence, the Zero-Bid settlement anomaly, and the scrambled-prior degradation reflect a shared methodological vulnerability: standard evaluation protocols test physics-constrained models without distinguishing physical inductive bias from non-specific numerical smoothing, software execution fidelity from published equations, or forecasting skill from market settlement structures. Auditing constraint provenance, software execution, telemetry latency, and coupled two-settlement cashflows provides the empirical

foundation necessary to determine when domain physical priors deliver genuine operational value in power systems.

Physics-constrained architectures offer tangible inductive value under data scarcity, but their operational benefits require rigorous empirical boundaries. Establishing those boundaries demands parameter-isolated ablations, scrambled negative controls, static computational graph verification, and market evaluation protocols that separate forward commitment revenue from imbalance settlement cashflow.

10. Open Data, Software Codebases, and Experimental Specifications

10.1. Summary of Resources and Infrastructure

10.2. Data and Code Availability

Raw operational SCADA telemetry from the Kelmarsh (6 turbines, 12.3 MW, Senvion MM92) and Penmanshiel (14 turbines, 28.7 MW, Senvion MM82) wind farms is openly available on Zenodo under CC-BY 4.0 licenses from Cubico Sustainable Investments (DOIs: 10.5281/zenodo.5841834 and 10.5281/zenodo.5946808). Operational SCADA telemetry from the Hill of Towie wind farm (21 turbines, 48.3 MW, Siemens SWT-2.3-82) is openly available on Zenodo under CC-BY 4.0 license released by RES on behalf of The Renewables Infrastructure Group (TRIG) (DOI: 10.5281/zenodo.20204946). Elexon System Buy Price (SBP) and System Sell Price (SSP) records are freely accessible through the Elexon BMRS portal under the UK Open Government Licence v3.0. The complete, reproducible Python verification suite—including computational graph static parsers, 2,000-replication circular block bootstrap scripts, Elexon BSC physical settlement calculators, and figure reproduction scripts—is openly archived at Zenodo and GitHub (`anonymous-audit/pcwpf-audit`). The full 46-study scoping extraction database with verbatim textual quotes, equation numbers, and multi-coder Fleiss' κ consensus scores is provided as open supplementary data.

Further technical information, trained model weights, or raw inter-rater reliability extraction matrices will be fulfilled by the corresponding author, Junyu Li (`ljylikezmn999@gmail.com`).

10.3. Wind Farm Telemetry and Data Sanitation Protocols

This computational study utilized field telemetry records from three commercial utility-scale wind farms located across the United Kingdom. The **Kelmarsh Wind Farm** (Northamptonshire, England) represents flat open inland topography and comprises six Senvion M-M92 turbines (rated 2.05 MW each, hub height 78 m, rotor diameter 92 m, farm capacity 12.3 MW), with training observations spanning $N_{\text{train}} = 5,098$ clean 10-minute intervals. The **Hill of Towie Wind Farm** (Drummuir, Scotland) represents complex upland terrain featuring micro-topographical wake interactions, comprising 21 Siemens SWT-2.3-82 turbines (rated 2.3 MW each, hub height 60 m, rotor diameter 82 m, farm capacity 48.3 MW), evaluated as a full-year independent spatiotemporal extrapolation benchmark with 7,599 forecast origins (45,594 forward settlement half-hours in 2017). The **Penmanshiel Wind Farm** (Scottish Borders) represents undulating upland terrain comprising 14 Senvion MM82 turbines (rated 2.05 MW each, farm capacity 28.7 MW), configured with a restricted training partition ($N_{\text{train}} = 1,583$ intervals) specifically to evaluate overfitting and prior inductive bias under sample-restricted regimes (acknowledging that site topography and turbine models co-vary with sample size across wind farms). Standard data sanitation protocols removed periods of turbine curtailment, anemometer icing, and communication maintenance dropouts ($P \leq 0$ when $v > v_{\text{cut-in}}$ or active power frozen for > 6 consecutive steps).

10.4. Method Details and Statistical Estimation

Systematic Scoping Review Protocol (PRISMA-ScR). Literature identification followed a structured protocol specified prior to full-text extraction, executing four Boolean title queries via the OpenAlex API (Q_1 through Q_4), complemented by manual keyword-directed discovery across Google Scholar, arXiv, and preprint servers for papers published between January 2023 and March 2026 (with rolling monitoring through July 2026), yielding 89 raw candidate records. Two-phase screening (title/abstract followed by full-text extraction) identified our core scoping cohort of 46 contemporary studies (38 peer-reviewed journal articles and 8 preprints, 2023–2026) subjected to exhaustive five-dimensional ($\mathcal{D}_1$–$\mathcal{D}_5$) extraction and inter-rater reliability coding. A targeted sensitivity search expanding retrieval to title, abstract, and author keywords across Scopus and IEEE Xplore (encompassing broader descriptors such as `"knowledge-guided"`, `"hybrid"`, and `"domain-informed"`) retrieved 14 additional candidate papers, with full-text inspection confirming zero instances of documented operational failure boundaries, specifically stress-testing the 0.0% negative trade-off finding. Code availability and ablation metrics (95.7% open-code deficit and 50.0% confounded module ablations) are strictly bounded to the audited 46-study cohort with Wilson confidence intervals. Inter-rater reliability

Table 8: Summary of Open Datasets, Audited Software Codebases, and Hardware Infrastructure.

Category	Resource / Data Source	Identifier / Access Link
Commercial Wind Farm Telemetry and Market Clearing Records		
Kelmarsh Wind Farm 10-min SCADA	Cubico Sustainable Investments / Zenodo	DOI: 10.5281/zenodo.5841834
Penmanshiel Wind Farm 10-min SCADA	Cubico Sustainable Investments / Zenodo	DOI: 10.5281/zenodo.5946808
Hill of Towie Wind Farm 10-min SCADA	TRIG / RES / Zenodo (Open Data)	DOI: 10.5281/zenodo.20204946; Elexon BM Unit: `T_HILW-1`
UK Elexon BMRS Market Clearing Prices	Elexon Balancing and Settlement Code	Open Government Licence v3.0
Systematic Scoping Audit Corpus ($N = 46$)	This Paper (Verbatim Extraction Ledger)	Data Repository: `corpus48_consensus`
Open-Source Software Implementations and Audit Scripts		
Python Programming Language (v3.10)	Python Software Foundation	RRID: SCR_008394
PyTorch Deep Learning Library (v2.0+)	PyTorch Foundation	RRID: SCR_018536
Statsmodels (Circular Block Bootstrap)	Statsmodels Developers	RRID: SCR_016074
Scikit-learn (Machine Learning in Python)	Pedregosa et al., 2011	RRID: SCR_002577
DLinear Baseline Codebase	Zeng et al., AAAI 2023	GitHub: `cure-lab/LTSF-Linear`
PatchTST Benchmark Codebase	Nie et al., ICLR 2023	GitHub: `yuqinie98/PatchTST`
Audited PCT Codebase	Wang et al., *Sci. Rep.* 2026	GitHub: `wind-power-forecasting/PCT`
Audited PINN-UQ Codebase	Xu et al., *Eng. App. AI* 2025	GitHub: `xuchen/PINN-UQ`
Full Empirical Audit and Verification Suite	This Paper	GitHub: anonymous-audit/pcwpf-audit
High-Performance Computing Hardware		
NVIDIA RTX 4090 Dedicated GPUs (10×)	NVIDIA Corporation	Compute Capability 8.9 (24GB VRAM each)

was benchmarked across a representative 20-study pilot by three independent raters (Coders A, B, and C) achieving Fleiss' $\kappa = 0.646$–1.000 across provenance, stage, and telemetry dimensions.

Static Code and Computational Graph Audit. Software codebases were audited via static AST parsing and line-by-line algebraic comparison against published manuscripts. For PCT (*Sci. Rep.* 2026), the loss definition in `tst/loss.py` was decompiled into symbolic PyTorch operations, establishing that `F.kl_div(p.log(), q)` computes $\sum q_i \log(q_i/p_i) = \mathrm{KL}(q \parallel p)$ instead of the published $\mathrm{KL}(p \parallel q)$. Algebraic tensor expansion of the empirical density kernel $K_p^\top K_w$ verified strict equivalence to a 2D product kernel density estimator.

10-GPU Prospective Controlled Ablation Testbed. Ablation experiments were executed in parallel across a dedicated cluster of 10 NVIDIA RTX 4090 GPUs distributed across two nodes: Node 1 (4 GPUs) evaluated Kelmarsh across 4 random initialization seeds (42, 43, 44, 45); Node 2 (6 GPUs) evaluated Hill of Towie and Penmanshiel across 3 random seeds each (42, 43, 44). All models were implemented in PyTorch under CUDA. Neural backbones were trained using Adam with learning rate 10^{-3} and batch size 128 in accordance with the frozen benchmark protocol. Regularization weight λ was evaluated across $\{0.01, 0.1, 1.0\}$. For pseudo-scrambled controls, the empirical wind-to-power mapping was permuted along the wind axis, preserving identical loss formulas, parameter counts, and gradient scales. Importantly, the primary forecasting models (DLinear) take exclusively historical power sequences ($x[:, :, 0]$) as inputs at inference, requiring zero runtime meteorological telemetry. The historical wind speed proxy lag scan ($k \in \{6, 12, 18, 36\}$ steps, 1 h to 6 h) evaluates the sensitivity of the training loss objective when indexing offline power curve priors via lagged historical wind speed measurements in the absence of forward numerical weather predictions.

Quantification and Statistical Analysis

Operational Decision Risk Formulation. To eliminate symmetric evaluation bias, performance was quantified using the non-negative asymmetric Pinball Loss function:

$$\mathcal{L}_q(y, \hat{y}) = \max\left(q(y - \hat{y}),\ (1 - q)(\hat{y} - y)\right), \tag{28}$$

where $q = \frac{1}{r+1} = \frac{1}{3}$ corresponds to an operational market penalty ratio $r = 2$ (under-generation penalty twice as costly as spill). Total normalized static cost was defined as $\text{Cost}_{\text{static}} = 3 \times \mathcal{L}_{1/3}(y, \hat{y})$, with validation-set quantile post-hoc offset calibration $\delta^* = \arg\min_\delta \hat{\mathcal{L}}_{1/3}(y, \hat{y} + \delta)$.

Circular Block Paired Bootstrap Inference. Statistical significance of exploratory dynamic price model contrasts ($\Delta\text{Cost} = \text{Cost}_{\text{method}} - \text{Cost}_{\text{baseline}}$, Table 4) was evaluated via circular block paired bootstrapping with 2,000 replications. Block length was set to $L = 1,008$ intervals (7 days) to rigorously preserve temporal autocorrelation and diurnal meteorological cycles. Two-sided 95% percentile confidence intervals were constructed from the empirical bootstrap distribution, rejecting the null hypothesis of equal performance if and only if zero fell outside the confidence interval ($p < 0.05$). For the prospective 3D controlled ablations in Table 3, metric margins represent sample means and standard deviations across independent random initialization seeds.

Appendix A. Mathematical Analysis of Decoupled Calibration Boundaries and Counterexamples

This appendix provides formal mathematical proofs for the two boundary counterexamples presented in Section 7, delineating the theoretical limits of decoupled decision calibration.

Appendix A.1. Counterexample 1: Constant Shift Invariance and Zero Excess Risk

Counterexample 1 (Constant Shift Invariance). Let the target variable follow a uniform distribution $Y \sim U[0.25, 0.75]$, and consider an asymmetric cost ratio $r = 2$ ($\tau = 1/3$). Suppose an unconstrained model outputs the exact conditional expectation $f_0 = \mathbb{E}[Y] = 0.50$, while a physics-constrained model incurs a constant positive shrinkage shift $c = +0.05$, outputting $f_\lambda = 0.55$.

Prediction Representation Error (MSE). The mean squared prediction errors are:

$$\text{MSE}(f_0) = \text{Var}(Y) = \frac{0.50^2}{12} \approx 0.020833, \tag{A.1}$$

$$\text{MSE}(f_\lambda) = \text{MSE}(f_0) + c^2 \approx 0.023333. \tag{A.2}$$

The physical constraint strictly inflates prediction MSE by $\Delta\text{MSE} = +0.0025$ (+12.0%).

Uncalibrated Operational Dispatch Loss. Under uncalibrated dispatch ($p = f$), the expected asymmetric loss is:

$$\begin{aligned}\mathcal{R}_{\text{uncal}}(p) &= 2\int_{0.25}^{p} (p - y) \cdot 2\, dy + \int_{p}^{0.75} (y - p) \cdot 2\, dy \\ &= 2(p - 0.25)^2 + (0.75 - p)^2.\end{aligned} \tag{A.3}$$

Evaluating at each prediction:

$$\mathcal{R}_{\text{uncal}}(f_0 = 0.50) = 2(0.25)^2 + (0.25)^2 = 0.187500, \tag{A.4}$$

$$\mathcal{R}_{\text{uncal}}(f_\lambda = 0.55) = 2(0.30)^2 + (0.20)^2 = 0.220000. \tag{A.5}$$

Post-Hoc Decoupled Quantile Calibration. The theoretical Bayes-optimal quantile at $\tau = 1/3$ is:

$$q_{1/3} = 0.25 + \tfrac{1}{3}(0.75 - 0.25) \approx 0.416667. \tag{A.6}$$

Under decoupled quantile calibration, each model searches for its optimal scalar displacement:

$$\delta_0^* = q_{1/3} - f_0 \approx 0.416667 - 0.500000 = -0.083333, \tag{A.7}$$

$$\delta_\lambda^* = q_{1/3} - f_\lambda \approx 0.416667 - 0.550000 = -0.133333. \tag{A.8}$$

Notice that the resulting dispatch commitments are identical:

$$p_0^* = f_0 + \delta_0^* = 0.416667, \quad p_\lambda^* = f_\lambda + \delta_\lambda^* = 0.416667. \tag{A.9}$$

Consequently, the post-calibration operational risks are identical:

$$\mathcal{R}_{\text{cal}}(p_0^*) = \mathcal{R}_{\text{cal}}(p_\lambda^*) = 0.166667 \implies \Delta\mathcal{R}_{\text{disp}} \equiv 0. \tag{A.10}$$

Theoretical Takeaway. Counterexample 1 proves that **an increase in prediction MSE is compatible with invariant post-calibration operational risk**. When constraint-induced representation error consists of a uniform shift, scalar calibration absorbs the distortion entirely, establishing that strict positive excess risk lower bounds cannot hold unconditionally.

Appendix A.2. Counterexample 2: Spatial Heteroscedasticity and Risk Reversal

Counterexample 2 (Spatial Heteroscedasticity). Let $X \in \{0, 1\}$ be a binary indicator with $P(X = 0) = P(X =$

1) = 0.5. Let the conditional target distributions exhibit heteroscedastic spread:

$$Y \mid (X = 0) \sim U[0.20, 0.60] \quad (\mu_0 = 0.40, s_0 = 0.40), \tag{A.11}$$

$$Y \mid (X = 1) \sim U[0.30, 0.90] \quad (\mu_1 = 0.60, s_1 = 0.60). \tag{A.12}$$

Let the asymmetric cost ratio be $r = 2$ ($\tau = 1/3$). Consider two candidate forecasters: Model A (the unbiased conditional mean) with predictions $f_A(0) = 0.40$ and $f_A(1) = 0.60$; and Model B (a biased estimator with higher MSE) with predictions $f_B(0) = 0.40$ and $f_B(1) = 0.55$.

Prediction MSE Comparison. The mean squared prediction errors are:

$$\begin{aligned} \mathrm{MSE}(f_A) &= 0.5\left(\frac{0.40^2}{12}\right) + 0.5\left(\frac{0.60^2}{12}\right) \\ &= \frac{13}{600} \approx 0.021667, \end{aligned} \tag{A.13}$$

$$\begin{aligned} \mathrm{MSE}(f_B) &= \mathrm{MSE}(f_A) + 0.5(0.55 - 0.60)^2 \\ &= 0.021667 + 0.001250 \approx 0.022917. \end{aligned} \tag{A.14}$$

Model B has strictly higher prediction MSE by $\Delta\mathrm{MSE} = +0.001250$ (+5.8%).

Post-Hoc Global Scalar Calibration. The local Bayes-optimal quantiles at $\tau = 1/3$ are:

$$q_{1/3}(0) = 0.20 + \tfrac{0.40}{3} = \tfrac{1}{3} \approx 0.333333, \tag{A.15}$$

$$q_{1/3}(1) = 0.30 + \tfrac{0.60}{3} = 0.500000. \tag{A.16}$$

The Bayes-optimal operational dispatch loss is $\mathcal{R}(q_\tau) = 0.5(\frac{0.40}{3}) + 0.5(\frac{0.60}{3}) = \frac{1}{6} \approx 0.166667$. The local conditional densities are $f_0(y) = 2.5$ and $f_1(y) = 1.6667$.

Because local quantile-to-mean shifts differ ($\Delta_0 = q_{1/3}(0) - \mu_0 = -\frac{1}{15} \approx -0.066667$ versus $\Delta_1 = q_{1/3}(1) - \mu_1 = -0.100000$), a single global scalar offset δ^* cannot simultaneously achieve both local quantiles. The expected operational risk under global shift δ is:

$$\mathcal{R}(f + \delta) = 0.5\,\mathcal{R}_0(f(0) + \delta) + 0.5\,\mathcal{R}_1(f(1) + \delta). \tag{A.17}$$

For **Model A** ($f_A = \mu$), the first-order optimality condition for δ_A^* is:

$$(r + 1)\left[0.5 \cdot 2.5(\delta_A^* - \Delta_0) + 0.5 \cdot 1.6667(\delta_A^* - \Delta_1)\right] = 0, \tag{A.18}$$

yielding the density-weighted optimal global offset:

$$\begin{aligned} \delta_A^* &= \frac{2.5(-\frac{1}{15}) + 1.6667(-0.10)}{2.5 + 1.6667} \\ &= \frac{-0.333333}{4.166667} = -0.080000. \end{aligned} \tag{A.19}$$

The resulting excess operational risk for Model A is:

$$\begin{aligned} &\mathcal{R}(f_A + \delta_A^*) - \mathcal{R}(q_\tau) \\ &= \frac{3}{2}\Big[1.25(-0.080 + \tfrac{1}{15})^2 \\ &\qquad + 0.8333(-0.080 + 0.100)^2\Big] \\ &= \frac{15}{8}\left(-\frac{1}{75}\right)^2 + \frac{5}{4}\left(\frac{1}{50}\right)^2 \\ &= \frac{1}{3000} + \frac{1}{2000} = \frac{1}{1200} \approx 0.000833, \end{aligned} \tag{A.20}$$

giving an absolute operational dispatch risk of:

$$\mathcal{R}_{\mathrm{cal}}(f_A + \delta_A^*) = \frac{1}{6} + \frac{1}{1200} = \frac{201}{1200} = 0.167500. \tag{A.21}$$

For **Model B** ($f_B(0) = 0.40$, $f_B(1) = 0.55$), the downward bias occurs on the higher-variance regime ($X = 1$). The required local offsets become:

$$\Delta_{0,B} = q_{1/3}(0) - 0.40 = -\tfrac{1}{15} \approx -0.066667, \tag{A.22}$$

$$\Delta_{1,B} = q_{1/3}(1) - 0.55 = 0.50 - 0.55 = -0.050000. \tag{A.23}$$

Notice that the discrepancy between regime offsets is halved: $|\Delta_{1,B} - \Delta_{0,B}| = \frac{1}{60} \approx 0.016667$, compared to $\frac{1}{30} \approx 0.033333$ for Model A. The optimal global offset for Model B is:

$$\begin{aligned} \delta_B^* &= \frac{2.5(-\frac{1}{15}) + 1.6667(-0.050)}{2.5 + 1.6667} \\ &= \frac{-0.166667 - 0.083333}{4.166667} = -0.060000. \end{aligned} \tag{A.24}$$

The resulting excess operational risk for Model B is:

$$\begin{aligned} &\mathcal{R}(f_B + \delta_B^*) - \mathcal{R}(q_\tau) \\ &= \frac{3}{2}\Big[1.25(-0.060 + \tfrac{1}{15})^2 \\ &\qquad + 0.8333(-0.060 + 0.050)^2\Big] \\ &= \frac{15}{8}\left(\frac{1}{150}\right)^2 + \frac{5}{4}\left(-\frac{1}{100}\right)^2 \\ &= \frac{1}{12000} + \frac{1}{8000} = \frac{1}{4800} \approx 0.000208, \end{aligned} \tag{A.25}$$

yielding an absolute operational dispatch risk of:

$$\mathcal{R}_{\mathrm{cal}}(f_B + \delta_B^*) = \frac{1}{6} + \frac{1}{4800} = \frac{801}{4800} = 0.166875. \tag{A.26}$$

Comparing the two post-calibration dispatch losses:

$$\mathcal{R}_{\mathrm{cal}}(f_B + \delta_B^*) = 0.166875 < 0.167500 = \mathcal{R}_{\mathrm{cal}}(f_A + \delta_A^*). \tag{A.27}$$

Model B achieves a 75% reduction in excess risk (0.000208 vs. 0.000833) and strictly lower operational dispatch loss than the unbiased conditional mean, despite exhibiting +5.8% higher prediction MSE.

Theoretical Takeaway. Counterexample 2 rigorously proves that **under conditional heteroscedasticity, a biased predictor with higher MSE can achieve strictly lower post-calibration dispatch loss than the true conditional mean**. This formalizes why the performance of physical constraints under decoupled calibration depends intricately on local error variance and environmental terrain.

Declaration of Interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Author Contributions (CRediT)

Junyu Li: Conceptualization, Methodology, Software, Formal analysis, Investigation, Data curation, Visualization, Project administration, Writing – original draft, Writing – review & editing. **Juntao Du**: Validation, Data curation, Software (Inter-rater reliability arbitration), Investigation, Writing – review & editing.

Declaration of Generative AI in the Writing Process

During the preparation of this manuscript, the authors utilized large language model assistants strictly for grammar checking, code syntax auditing, and bibliographic formatting. After using these tools, the authors reviewed and edited the content thoroughly and take full responsibility for the scientific integrity, analytical claims, and conclusions presented in this publication.

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References

[1] H. Holttinen, et al., Design and operation of power systems with large amounts of wind power, Tech. rep., IEA Wind TCP Task 25 (2021).

[2] R. Weron, Electricity price forecasting: A review of the state-of-the-art, International Journal of Forecasting 30 (4) (2014) 1030–1081.

[3] Z. C. Lipton, J. Steinhardt, Troubling trends in machine learning scholarship, Communications of the ACM 62 (6) (2019) 45–53.

[4] P. Pinson, C. Chevallier, G. N. Kariniotakis, Trading wind generation from short-term probabilistic forecasts—coordinated electricity market, IEEE Transactions on Power Systems 22 (3) (2007) 1148–1156. doi:10.1109/TPWRS.2007.901611.

[5] M. Raissi, P. Perdikaris, G. E. Karniadakis, Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations, Journal of Computational Physics 378 (2019) 686–707.

[6] G. E. Karniadakis, I. G. Kevrekidis, L. Lu, P. Perdikaris, S. Wang, L. Yang, Physics-informed machine learning, Nature Reviews Physics 3 (6) (2021) 422–440.

[7] L. Sheng, C. Li, M. Gao, X. Xi, D. Zhou, A review of SCADA-based condition monitoring for wind turbines via artificial neural networks, Neurocomputing 633 (2025) 129830. doi:10.1016/j.neucom.2025.129830.

[8] D. Wang, Q. Luo, J. Gao, D. Zhang, Y. Chen, Physics-constrained transformer for wind power forecasting, Scientific Reports 16 (1) (2026) 4237. doi:10.1038/s41598-025-34348-x.

[9] S. Kapoor, A. Narayanan, Leakage and the reproducibility crisis in machine-learning-based science, Patterns 4 (9) (2023) 100804.

[10] J. Pineau, P. Vincent-Lamarre, K. Sinha, V. Larivi
'ere, A. Beygelzimer, F. d'Alch
'e Buc, E. Fox, H. Larochelle, Improving reproducibility in machine learning research: A report from the NeurIPS 2019 reproducibility program, Journal of Machine Learning Research 22 (164) (2021) 1–20.

[11] J. Gao, Y. Cheng, D. Zhang, Y. Chen, Physics-constrained wind power forecasting aligned with probability distributions for noise-resilient deep learning, Applied Energy 383 (2025) 125295. doi:10.1016/j.apenergy.2025.125295.

[12] N. Zehtabiyan-Rezaie, A. Iosifidis, M. Abkar, Physics-guided machine learning for wind-farm power prediction: Toward interpretability and generalizability, PRX Energy 2 (1) (2023) 013009. doi:10.1103/PRXEnergy.2.013009.

[13] A. Krishnapriyan, A. Gholami, S. Zhe, R. Kirby, M. W. Mahoney, Characterizing possible failure modes in physics-informed neural networks, in: Advances in Neural Information Processing Systems, Vol. 34, 2021, pp. 26548–26560.

[14] H. Madsen, P. Pinson, G. Kariniotakis, T. S. Nielsen, H. A. Nielsen, Standardizing the performance evalua-

tion of short-term wind power prediction models, Wind Engineering 29 (6) (2005) 475–489.

[15] M. Optis, N. Bodini, S. Redfern, J. Fang, Evaluating wind speed and power forecasts using WE-Validate: An open-source and systematic validation framework, Renewable Energy 199 (2022) 1577–1589.

[16] Y. Zhang, J. Wang, X. Wang, Review on probabilistic forecasting of wind power generation, Renewable and Sustainable Energy Reviews 32 (2014) 255–270.

[17] P. B. Dao, W. J. Staszewski, T. Barszcz, T. Uhl, Condition monitoring and fault detection in wind turbines based on cointegration analysis of SCADA data, Renewable Energy 116 (2018) 107–122. doi:10.1016/j.renene.2017.06.089.

[18] J. Yan, X. Li, H. Wang, F. Si, W. Qiao, Y. Liu, The economic value of wind power forecasting: a data-driven method and its application in various scenarios, Energy 340 (2025) 139151. doi:10.1016/j.energy.2025.139151.

[19] A. D'Amour, K. Heller, D. Moldovan, B. Adlam, I. Alabdulmohsin, L. E. Celis, J. Frankle, C. George, M. Hobson, A. Hu, et al., Underspecification presents challenges for credibility in modern machine learning, Journal of Machine Learning Research 23 (2022) 1–61.

[20] O. E. Gundersen, S. Kjensmo, State of the art: Reproducibility in artificial intelligence, in: Proceedings of the AAAI Conference on Artificial Intelligence, Vol. 32, 2018, pp. 1644–1651.

[21] J. Yan, Y. Liu, S. Han, Y. Wang, S. Feng, Reviews on wind speed and wind power forecasting methods, Renewable and Sustainable Energy Reviews 58 (2016) 429–441.

[22] J. Maldonado-Correa, S. Martín-Martínez, E. Artigao, E. Gómez-Lázaro, Using SCADA data for wind turbine condition monitoring: A systematic literature review, Energies 13 (12) (2020) 3132. doi:10.3390/en13123132.

[23] A. C. Tricco, E. Lillie, W. Zarin, K. K. O'Brien, H. Colquhoun, D. Levac, D. Moher, M. D. Peters, T. Horsley, L. Weeks, et al., PRISMA extension for scoping reviews (PRISMA-ScR): checklist and explanation, Annals of Internal Medicine 169 (7) (2018) 467–473.

[24] D. Levac, H. Colquhoun, K. K. O'Brien, Scoping studies: advancing the methodology, Implementation Science 5 (1) (2010) 69.

[25] R. Xu, T. Wu, G. Zhang, Y. Li, C. Guo, B. Yang, Y. Li, Uniwind: Toward unified day-ahead wind power forecasting via physics-informed state routing, arXiv preprint arXiv:2607.01670 (2026).

[26] Renewable Energy Systems, The Renewables Infrastructure Group, Hill of Towie Wind Farm Open SCADA Data (2025). doi:10.5281/zenodo.20204946.
URL https://doi.org/10.5281/zenodo.20204946

[27] Cubico Sustainable Investments Ltd, Kelmarsh Wind Farm Open SCADA Data (2021). doi:10.5281/zenodo.5841834.
URL https://doi.org/10.5281/zenodo.5841834

[28] Cubico Sustainable Investments Ltd, Penmanshiel Wind Farm Open SCADA Data (2022). doi:10.5281/zenodo.5946808.
URL https://doi.org/10.5281/zenodo.5946808

[29] Y. Nie, N. H. Nguyen, P. Sinthong, J. Kalagnanam, A time series is worth 64 words: Long-term forecasting with transformers, in: The Eleventh International Conference on Learning Representations (ICLR 2023), 2023, pp. 1–15.
URL https://openreview.net/forum?id=Jbdc0vTOcol

[30] C. Xu, J. Wang, L. Zhang, Y. Liu, Physics-informed neural networks with uncertainty quantification for wind turbine power modeling, Engineering Applications of Artificial Intelligence 140 (2025) 109650. doi:10.1016/j.engappai.2024.109650.

[31] R. Koenker, Quantile Regression, Cambridge University Press, 2005. doi:10.1017/CBO9780511754098.

[32] S. Kapoor, E. Cantrell, K. Peng, T. D. Pham, C. A. Bail, M. De-Arteaga, J. Frankle, M. Hardt, T. Hashimoto, P. Henderson, F. Khani, Z. C. Lipton, A. Narayanan, REFORMS: Reporting standards for machine learning based science, Science 384 (6696) (2024) 624–626. doi:10.1126/science.adk3452.

[33] I. Walsh, D. Fishman, D. Garcia-Gasulla, M. Titov, G. Pollastri, J. Harrow, F. Psomopoulos, S. C. E. Tosatto, DOME: recommendations for supervised machine learning validation in biology and medicine, Nature Methods 18 (10) (2021) 1122–1127. doi:10.1038/s41592-021-01205-4.